A MODERN LOOK AT CLASSICAL GENERALIZATION BOUNDS

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Abstract.

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1. Introduction

In machine learning, there exists a large number of algorithms, notably neural networks trained with gradient descent, which achieve great performances at prediction tasks on unseen data. A big question in learning theory is explaining what drives such *generalization* [19, 16]. Generalization is the process of applying a model to unseen data, after the algorithm has learned it from some available data. A number of results have tried to answer this question, coming to the apparent conclusion that the classical theory is not able to exhaustively explain what drives the generalization ability of learning algorithms. In particular, it fails to describe the behavior of the test error of learning algorithms as the complexity of the hypotheses set increases [3, 19, 20, 15].

This thesis seeks to understand whether this is true by testing theoretical results against empirical evidences. We study the setting of a finite hypotheses set of binary decision trees. We look for a bound on the *generalization gap* as the maximum depth of decision trees increases. The generalization gap is defined as the expected difference between the expected and empirical risk of the selected model,

$$\mathbb{E}_D \left[(R - \hat{R}_D)(\hat{h}_D) \right]$$

where $\mathbb{E}_D$ denotes the expectation over the disorder of the data.

We consider a number of candidates coming from existing theoretical results that increasingly account for the structure of the learning problem at hand, in particular the finiteness of decision trees and the noise in the data. Moreover, we test the validity of these results comparing them with the empirical test error curve by carrying out experiments on synthetic data. We are interested in capturing the qualitative behavior of the test error curve. This means that we are not concerned with finding a quantitatively correct bound that lies above the generalization gap, as long as it reproduces its shape. By doing so, it is possible to examine how the quantities appearing in the bounds affect their shape, and thereby identify which aspects of the learning problem influence the generalization ability of decision trees.

The final result of this thesis is a qualitative bound on the generalization gap of decision trees that is empirically validated, providing a theoretical understanding

of the behavior of decision trees on unseen data. It also develops framework and toolkit that can be extended to the study of the generalization ability of other learning algorithms, like gradient-based ones.

1.1. **Previous work.** This section presents the state of the art and how the present work relates to it.

The behaviour of the test error of tree-based models as a function of their complexity has itself been the object of debate. [3] report a double descent curve for tree-based methods like random forests and boosting, where the test error decreases again beyond the interpolation threshold. [6] show that this second descent arises from combining distinct complexity axes into a single one, such as the number of leaves of a tree and the number of trees in an ensemble, and that the curves recover a traditional convex shape when plotted against an effective number of parameters. In our setting, complexity is measured along a single axis, the maximum depth d of a single decision tree. Along this axis, once the learned tree interpolates the training sample, a larger depth no longer changes the learned model, and the empirical test error saturates (Section 2); a second descent can only arise by moving to a different complexity axis, such as the number of trees in an ensemble. In this thesis, we limit the analysis to decision trees and not ensembles of these.

Several works evaluate generalization bounds for decision trees in practice, but with a motivation different from ours. Specifically, [10] use Rademacher complexity as an empirical complexity measure: they compute the Rademacher penalty $\sup_{\ell_h \in \mathscr{L}} \left| \frac{1}{n} \sum_i \xi_i \ell_h(X_i, Y_i) \right|$ on a single realization of D and ξ, and plug it into a high-probability bound on $R(h)$. Their hypotheses set is data-dependent, namely the prunings of a tree grown on a separate sample, for which reduced error pruning applied to relabeled data computes the supremum exactly. They compare this bound with an Occam's razor bound based on a uniform code length over the prunings, analogous to our Theorem 2.6. Our work differs in that we consider hypotheses sets $\mathscr{H}_d$ fixed in advance and aim at interpretable analytical expressions rather than numerical values; hence, we use $\mathcal{R}_n(\mathscr{L})$ only as an intermediate step towards $N_D(\mathscr{H})$.

In [12], the authors derive an analytical upper bound on the growth function of classes of binary decision trees with fixed structure, that is, a structure-dependent count of the distinct classifications that such trees can realize on a sample, more detailed than the counting inequality $\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$ of Section 5. They use it within a structural risk minimization bound as a pruning criterion, evaluated through the test accuracy of the pruned trees. Our interest is different: rather than using a bound for model selection, we compare its shape with the empirical generalization gap as a function of the complexity of the hypotheses set.

Other works compare the numerical tightness of several generalization bounds on benchmark datasets [9, 17], or use bounds on the VC dimension of decision trees for pruning [18]. Further papers propose bounds on the error of decision trees that depend on the specific training sample or on an effective number of leaves [13, 7, 2], without evaluating them empirically. Outside decision trees, [8] test a large number of complexity measures against the generalization of neural networks.

Although not specific to decision trees, [4] is a survey on statistical bounds on the risk of learning algorithms, and has been a fundamental guide for the theoretical side of this thesis. A difference, however, lies in the emphasis of the analysis. The survey is largely concerned with how bounds behave as the sample size n grows, as guarantees on the precision of the empirical risk in estimating the true risk. The same perspective underlies the statistical analyses of CART [11]. We, instead, keep n fixed and regard bounds as functions of the complexity of the hypotheses set, which should describe the qualitative behaviour of the generalization gap. For instance, our Theorem 4.2, analogous to Theorem 5.1 of [4], is crucial to this work because it allows us to include the noise in the bounds, whereas in [4] it is mainly used to improve the dependence on n when the training error goes to 0.

In light of the related literature, the contribution of this work is to study and empirically validate generalization bounds for decision trees with the aim of capturing the generalization behavior of the algorithm. To the best of our knowledge, this is the first work studying generalization bounds in this sense. As such, it develops a framework and approach that might be extended to the investigation of the generalization ability of other learning algorithms.

1.2. **Setting.** Given a sample space Ω, let inputs and labels be generated by $X : \Omega \to \mathcal{X} = \{\bar{x}_1, \dots, \bar{x}_N\}$ and $Y : \Omega \to \mathcal{Y} = \{-1, 1\}$. Let p be the joint probability distribution of (X, Y) on $\mathcal{X} \times \mathcal{Y}$, with

$$p_{j,y} = \Pr[X = \bar{x}_j, Y = y], \qquad p_{j,y} \geq 0, \qquad \sum_{j=1}^{N} \sum_{y \in \mathcal{Y}} p_{j,y} = 1.$$

Let $D = ((X_i, Y_i))_{i=1}^n$ be a dataset of n i.i.d. copies of (X, Y) drawn from p. We use the term *dataset* to refer to the *random variable* generating the data throughout this work. We denote by $S = (x_i, y_i)_{i=1}^n$ a realization of it. Given this data, we want to learn a classifier $h : \mathcal{X} \to \mathcal{Y}$ that given a new input $x \in \mathcal{X}$ classifies it correctly. Let $\mathscr{H} \subseteq \{h : \mathcal{X} \to \mathcal{Y}\}$ be a nonempty, finite set of functions where we look for such a classifier. Given a loss function $\ell : \mathbb{R} \times \mathcal{Y} \to \mathbb{R}^+$, let the true and empirical risks be

$$R(h) = \mathbb{E}_D(\ell(h(X), Y)) = \sum_{\bar{x} \in \mathcal{X}} \sum_{\bar{y} \in \{-1,1\}} \ell(h(\bar{x}), \bar{y}) p(\bar{x}, \bar{y}).$$

and

$$\hat{R}_D(h) = \frac{1}{n} \sum_{i=1}^{n} \ell(h(X_i), Y_i).$$

Because we do not have access to p, we select the *empirical risk minimizer* (ERM),

$$\hat{h}_D \in \arg\min_{h \in \mathscr{H}} \hat{R}_D(h)$$

which, importantly, is a random variable of the data D. Note that at least one such $\hat{h}_D$ exists, since $\mathscr{H}$ is finite and nonempty.

1.3. **Approach.** We study bounds on the expcted gap between the test and train error of the ERM,

$$\mathbb{E}_D \left[(R - \hat{R}_D)(\hat{h}_D) \right] \tag{1}$$

This approach allows to obtain bounds on the risk of the learnt model $\hat{h}_D$ instead of uniform bounds over the whole space of hypotheses, as the algorithm may not explore it [5, 15]. Moreover, the training error usually goes to 0 in our setting where the dataset is smaller than the input space, hence a bound on (1) corresponds to a bound on the test error for a large enough depth. In light of this, we refer to (1.2) as a bound on the gap and on the test error interchangeably throughout this work. We detach from the usual literature also in that we express concentration results as

bounds on the expected value of the target quantity via tail integration, rather than using results in high probability or bounding the probability that this quantity is large. In fact, results in expected value are more practical for the experiments and their interpretation is clearer as compared to high-probability results.

We state our initial results in a general setting, for arbitrary hypotheses classes and loss functions, while we develop our main results in the specific case of binary decision trees under the zero-one loss. Specifically,

- the zero-one loss is $\ell(h(x), y) = \mathbf{1}(h(x) \neq y) = \frac{1-h(x)y}{2}$;
- we consider a number of hypotheses classes $\mathscr{H}$, each denoted by $\mathscr{H}_d$ and corresponding to the set of binary decision trees $h : \mathcal{X} \to \mathcal{Y}$ with depth at most d for $d > 0$, letting d range over increasing values.

This is also the setting in which experiments are carried out. We test our theoretical findings by comparing them with empirical evidence. In doing so, we use the DecisionTreeClassifier algorithm from the scikit-learn Python library. This differs from the empirical risk minimization algorithm that appears in (1.2) in that it is a greedy algorithm, which not always selects the best possible model. Effects of this difference are discussed in Section 6.

Each one of Sections 2 to 5 is organized as follows. First, we present theoretical results in the most generic case given the setting above. Then, we describe the experimental design and state the final asymptotic bound, which we then plot alongside an empirical estimate of (1) to assess how well it captures the qualitative behavior of the test error curve. We conclude each section commenting on the results. Finally, in Section 6 we conclude and discuss limitations and potential developments of our findings.

All proofs, the preliminary experiments, some additional results, and code explanations are in Appendix A, B, **??**, and D respectively.

2. Union bound over $\mathscr{H}$

In this section, we present a fundamental result on the behavior of the gap between the expected and true risk of the ERM. First, we make the following assumption.

Assumption 2.1 (Boundedness of loss function)**.** *Given a set $A \subseteq \mathbb{R}$ and $\ell : A \to \mathbb{R}^+$, $\exists M \in \mathbb{R}, M > 0$ s.t.*

$$\ell(a) \leq M \qquad \forall a \in A.$$

A bounded loss function is crucial for obtaining all the upper bounds on the gap discussed in this work. Nonetheless, we now give a sufficient condition that allows to circumnavigate boundedness.

Assumption 2.2 (Lipschitzness of loss function and bound at zero)**.** *Given a set $A \subseteq \mathbb{R}$ and $\ell : A \to \mathbb{R}^+$,*

(i) $\exists C > 0$ *s.t.*

$$\forall a, a' \in A, |\ell(a) - \ell(a')| \leq C|a - a'|$$

(ii) $\exists C' > 0$ *s.t.*

$$|\ell(0)| \leq C'$$

Proposition 2.3. *Let $A = \{a \in \mathbb{R} : |a| \leq B$ for some constant $B \in \mathbb{R}\}$. Then Assumption 2.2 $\Rightarrow$ Assumption 2.1.*

Therefore, it is sufficient to assume Lipschitzness of the loss function to obtain boundedness, assuming bounded inputs.

On top of this, let $T : \mathbb{R} \to [-1, 1]$,

$$T(a) = \begin{cases} -1 & a \leq -1 \\ a & -1 < a < 1, \\ 1 & a \geq 1 \end{cases} \qquad a \in A \tag{2}$$

be a *truncation map*. For any hypotheses class $\mathscr{H} = \{h : \mathcal{X} \to \mathcal{Y}\}$, we can always define $\mathscr{H}_T = \{h_T \coloneqq T \circ h : \mathcal{X} \to [-1, 1] | h \in \mathcal{H}\}$ and take $A = \{(h(x), y) : h \in$

$\mathscr{H}_T, x \in \mathcal{X}, y \in \mathcal{Y} = \{1, -1\}\}$ which satisfies the assumption of bounded inputs required by Proposition 2.3.

Therefore, Proposition 2.3 and the truncation technique determine that Lipschitzness of the loss function immediately implies boundedness. The former is easier to verify, as Example 2.4 shows, and does not depend on the hypotheses set. This provides a framework for extending the results in this work to an infinte $\mathscr{H}$, as we discuss in Section 6.

Example 2.4 (Logistic loss function)**.** *Let $\ell(a) = \log(1+e^{-a})$, $a \in \mathbb{R}$ be the logistic loss function. Note that in our setting the function takes as input the product $h(x)\cdot y$. We have*

$$\ell'(a) = -\frac{e^{-a}}{1+e^{-a}} = -\frac{1}{1+e^{a}} \quad \Rightarrow \quad |\ell'(a)| < 1 \qquad \forall a \in \mathbb{R}$$

hence

$$C = \sup_{a\in\mathbb{R}} |\ell'| \leq 1.$$

Also,

$$\ell(0) = \log 2 = C'.$$

Fact 2.5. *The zero-one loss satisfies Assumption 2.1 with $M = 1$.*

We are ready to state our first result.

Theorem 2.6. *Let ℓ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$ by Proposition 2.3. Then*

$$0 \leq \mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_D)] \leq 2M\sqrt{\frac{\log|\mathscr{H}|}{2n}}$$

The logarithmic dependence on $|\mathscr{H}|$ arises directly from a union bound (Proposition A.4) applied over the hypotheses class. We establish the exponential concentration of $\hat{R}_D(h)$ for a single, fixed hypothesis $h \in \mathscr{H}$ using Hoeffding's inequality (Theorem A.1). However, the ERM $\hat{h}_D$ is a random variable dependent on the data D. To bound its deviation, we require the concentration event to hold simultaneously for all functions $h \in \mathscr{H}$. Taking the union bound introduces a factor of $|\mathscr{H}|$

into the probability tail. Finally, integrating this tail to bound the expected value yields the $\sqrt{\log|\mathscr{H}|}$ dependence.

This result formally demonstrates that the empirical risk of the ERM systematically underestimates its true risk, with the generalization gap scaling as $\mathcal{O}\left(\sqrt{\frac{\log|\mathscr{H}|}{n}}\right)$. We next present the experimental design to evaluate this bound for binary decision trees and compare its asymptotic behavior with the empirical test error curve.

Experimental design

We create the support $\mathcal{X}$ as a discretization over $N = 10,000$ numbers of the interval $[0, 10]$. We sample from it with the uniform measure $p_{\mathcal{X}}(\bar{x}) = \sum_{\bar{y}\in\mathcal{Y}} p(\bar{x}, \bar{y}) = \frac{1}{N}$. The law to generate the labels is

$$y \leftarrow \text{sign}(h^*(x) + \sigma\varepsilon)$$

where $h^* : \mathcal{X} \to \mathcal{Y}$, is the data-generating function, $\varepsilon \sim \mathcal{N}(0, 1)$, is a Gaussian noise, and $\sigma > 0$. First, we generate the labels with a centered parabola, so that roughly half of them are 1 and half -1 in the interval $[0, 10]$. Since an algorithm could learn such a function with few datapoints (see preliminary experiments in Appendix B.2), we switch to a sinusoid with varying frequency,

$$h^*(x) = 5\,\sin\left(\frac{\pi\,(m+1)}{10}\,x\right) \tag{3}$$

where m is the number of times h^* crosses the x-axis in the open interval $(0, 10)$, yielding $m + 1$ contiguous blocks of $1s$ and $-1s$. We set as default value $m = 10$.

We test a Bernoullian noise instead of a Gaussian one, concluding that there is no interesting difference in the performance of the algorithm as long as the portion of misclassified inputs due to the noise is the same (see preliminary experiments in Appendix B.1). We find that fixing $m = 10$, $\sigma = 3.5$ achieves the same noise as $p = 0.8$ (roughly 20% of labels flipped). Thus, we stick to a Gaussian noise.

All in all, our labels are generated as

$$y \leftarrow \text{sign}\left(5\,\sin\left(\frac{\pi\,(m+1)}{10}\,x\right) + \sigma\varepsilon\right)$$

with $\sigma = 3.5$ and $m = 10$. $\text{sign}(\cdot)$ maps 0 to 1.

We proceed as follows. We generate a test set with size $n_{\text{test}} = 200,000$ sampling from $\mathcal{X}$ with $p_{\mathcal{X}}$ and with replacement, and generating the labels as above. Then, for each depth $d \in [1, 7\lceil \log_2 n \rceil] \subset \mathbb{N}$ we do the following. For a number $T = 100$ of iterations, we

- generate a train set ($n = 50$);
- fit a decision tree with depth at most d (DecisionTreeClassifier algorithm from the scikit-learn library);
- compute the train error, the test error, and the differene between the two (the gap).

Then, we average the resulting quantities over the T samples, obtaining empirical estimates of the expected values of the train error, test error, and gap over the disorder of the data.

To plot the theoretical bound, instead, we use $|\mathscr{H}_d| = 2^d$.

As the maximum depth d increases, we plot the following:

1. the train and the test error curves;
2. the gap and the theoretical bound;
3. the test error and the sum of train error and the bound (moving $\hat{R}_D(\hat{h}_D)$ to the right).

Considering the zero-one loss, and ignoring constants since we are interested in the asymptotic behavior, Theorem 2.6 gives

$$\mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_{D,d})] \leq O\left(\sqrt{\frac{\log |\mathscr{H}_d|}{n}}\right) \qquad \text{for } d > 0. \tag{4}$$

We plot the bound (4) in Figure 1. The first thing catching the eye is that the bound diverges, while the test error saturates. The divergence is due to the dependence of the bound on $|\mathscr{H}_d|$, that is the number of binary trees with maximum depth d, which clearly increases with d. It follows immediately that a complexity measure of the hypotheses set that naively counts the functions it contains does not satisfy our requirements to find a qualitatively informative bound.

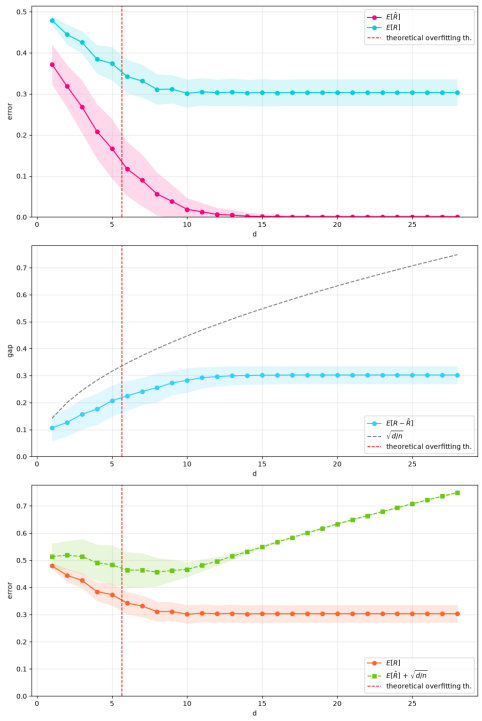


FIGURE 1. Union bound (Theorem 2.6) over $\mathscr{H}_d$. Train and test error, gap against the bound, and test error against train error plus the bound, as functions of the maximum depth d; $n = 50$, $m = 10$, Gaussian noise with $\sigma = 3.5$.

The test error of binary decision trees saturates because of a specific characteristic of the algorithm. As soon as the learned tree has only "pure" leaves, i.e. leaves containing only inputs that have the same label in the training set, the algortihm ceases to split them and hence any additional allowed depth does not change the learned model. Clearly, the cardinality of the hypotheses sets $|\mathscr{H}_d|$ does not capture this property.

Instead, a counting way that accounts for the finiteness of our problem might reproduce the saturation of the test error curve. Specifically, the configuration with only pure leaves can be achieved for two reasons: either all the possible values that the inputs can take are classified, or all the inputs in the sample are classified. Since in our case $n \ll N$, the second event happens much earlier than the first one. As

a consequence, a complexity measure that counts the number of distinct classifications that the functions in $\mathscr{H}$ can give, i.e. the vectors $h(x_1), \ldots, h(x_n) : h \in \mathscr{H}$, instead of the functions themselves, could more faithfully represent the true complexity of the space of models that the algorithm can practically learn and yield a bound that better captures the empirical test error curve.

3. Using the Rademacher complexity

In this section, we introduce the Rademacher complexity, a measure of expressiveness of a class of functions. This tool allows to leverage the viewpoint exposed in the last section and obtain a saturating bound on the test error of decision trees. As before, we state the first results for generic loss functions and hypotheses sets.

Definition 3.1 (Empirical Rademacher complexity)**.** *Let $\xi_1, \dots, \xi_n$ be n i.i.d. random variables taking value 1 or -1 with equal probability. Let $\mathcal{Z} \subset \mathbb{R}$ be a set and $\{z_1, \dots, z_n\} \subset \mathcal{Z}$ a collection of n points. Given $\mathcal{G} \coloneqq \{g : \mathcal{Z} \to \mathbb{R}\}$ a class of functions over $\mathcal{Z}$, the Rademacher complexity of $\mathcal{G}$ is*

$$\mathcal{R}_n(\mathcal{G}) \coloneqq \mathbb{E}_\xi \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^{n} \xi_i g(z_i) \right]$$

The Rademacher complexity measures the richness of a class of functions. The argument of the supremum $\frac{1}{n}\sum_i \xi_i g(z_i)$ is the (scaled) inner product between $(\xi_1, \dots, \xi_n)$ and $(g(z_1), \dots, g(z_n))$, measuring how well functions in $\mathcal{G}$ can align to the Rademacher variables.

In the context of our learning problem, we look at the Rademacher complexity of the class $\mathscr{L} = \{\ell_h : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^+ : \ell_h(x, y) = \ell(h(x), y), h \in \mathscr{H}\}$.

Definition 3.2 (Rademacher complexity of $\mathscr{L}$)**.** *Let $\xi = (\xi_1, \dots, \xi_n)$ be independent random variables taking value 1 or -1 with equal probability. The Rademacher complexity of the class $\mathscr{L}$ is*

$$\mathcal{R}_n(\mathscr{L}) \coloneqq \mathbb{E}_{\xi, D} \left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n} \sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i) \right]$$

Note that we also take the expectation over the disorder of the data D, because our sample $(X_1, Y_1), \dots, (X_n, Y_n)$ is random and our quantity of interest 1.2 is independent of the data realization. In this case, the argument of the supremum is the inner product between $(\xi_1, \dots, \xi_n)$ and $(\ell_h(X_1, Y_1), \dots, \ell_h(X_n, Y_n))$. Since the loss function is fixed, it measures to what extent classifiers $h \in \mathscr{H}$ can align to random noise. Note that the supremum is taken before the expectation over the disorder of ξ and D, and that it is a maximum, since $\mathscr{L}$ is finite. Hence, for

each dataset and for each realization of the Rademacher vectors, we can choose a different function $\ell_h \in \mathscr{L}$ that attains the maximum. A high complexity means that for most values of ξ we can find a model h that is similar to that pattern. It is the highest when we can find a function that correlates with any possible pattern.

The following results hold.

Theorem 3.3 (Rademacher complexity bound)**.** *[*[1]*, Proposition 4.2]*

$$\mathbb{E}_D\left[\sup_{h\in\mathscr{H}}(R-\hat{R}_D)(h)\right] \leq 2\mathcal{R}_n(\mathscr{L})$$

Lemma 3.4 (Massart)**.** *[*[14]*, Theorem 3.3] Let ℓ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$ by Proposition 2.3. Then*

$$\mathcal{R}_n(\mathscr{L}) \leq M\sqrt{\frac{2\log|\mathscr{L}|}{n}}.$$

Note that Theorem 3.3 can be easily extended to a bound on the gap at $\hat{h}_D$ because

$$\mathbb{E}_D[(R-\hat{R}_D)(\hat{h}_D)] \leq \mathbb{E}_D\left[\sup_{h\in\mathscr{H}}(R-\hat{R}_D)(h)\right]$$

since $\hat{h}_D \in \mathscr{H}$.

Remark 3.5. *We have $|\mathscr{L}| \leq |\mathscr{H}|$ by definition of $\mathscr{L}$.*

Altogether, the previous results yield a bound on the gap at the ERM that is asymptotically the same as that provided by Theorem 2.6. Now, we introduce some technology that allows to take a step forward.

Let

$$\mathscr{L}(D) \coloneqq \{(\ell_h(X_1,Y_1),\dots,\ell_h(X_n,Y_n)) : h \in \mathscr{H}, D = ((X_i,Y_i))_{i=1}^n\}$$

be the set of loss vectors for a random dataset D, i.e. the set containing the functions in $\mathscr{L}$ applied to D. Let $v_i = \ell_h(X_i,Y_i)$ to ease notation.

The supremum in Definition 3.2 can be equivalently computed over the set of loss vectors,

$$\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_i \xi_i\ell_h(X_i,Y_i) = \sup_{(v_1,\dots,v_n)\in\mathscr{L}(D)}\frac{1}{n}\sum_i \xi_i v_i$$

This yields the following version of the lemma.

Let

$$N_D(\mathscr{H}) \coloneqq |\{(\ell_h(X_1, Y_1), \dots, \ell_h(X_n, Y_n)) : h \in \mathscr{H}, D = ((X_i, Y_i))_{i=1}^n\}| = |\mathscr{L}(D)|$$

be the cardinality of the set of loss vectors for the hypotheses in *hyp*, which is a random variable as it depends on the dataset D.

Lemma 3.6 (Massart, loss vectors)**.** *Let ℓ be as in Assumption 2.2, so that $\ell(\cdot) \leq M$ for some $M \in \mathbb{R}$ by Proposition 2.3. Then*

$$\mathcal{R}_n(\mathscr{L}) \leq M\sqrt{\frac{2\log \mathbb{E}_D[N_D(\mathscr{H})]}{n}}$$

So, using the Rademacher complexity as an intermediate step allows us to replace the number of predictors $|\mathscr{H}|$ from Theorem 2.6 with the expected number of loss vectors $N_D(\mathscr{H})$ (up to constants). In fact, Theorem 3.3 bounds a quantity that depends on the true expected value of $\ell_h(X, Y)$ with a quantity that can be computed on loss vectors. The key ingredient that allows this is a symmetrization technique, which consists in replacing the deviation of the random variable $\ell_h(X, Y)$ from its expected value with the difference between two random variables $\ell_h(X, Y)$ and $\ell_h(X', Y')$ taking two i.i.d. copies D and D' as inputs. Note that since the loss function is fixed, counting loss vectors is equivalent to counting classification vectors. We work with loss vectors here because they arise more naturally from the Rademacher complexity, and turn to classification vectors in Section 5.

The following result encloses the key point of the saturation mechanism discussed at the end of Section 2.

Lemma 3.7 (Counting)**.** *Let ℓ be the zero-one loss. Then*

$$N_D(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$$

surely.

Substituting this bound on N_D in Lemma 3.6 and combining it with Theorem 3.3, we obtain the following bound.

Theorem 3.8 (Rademacher and Massart bound)**.** *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}_D[(R - \hat{R}_D)(\hat{h}_D)] \leq \min\left\{2\sqrt{\frac{2\log|\mathscr{H}|}{n}}, 2\sqrt{2\log 2}\right\}$$

The experimental design is the same as above. Again, we ignore the constants and use

$$\min\left\{2\sqrt{\frac{2\log|\mathscr{H}_d|}{n}}, 2\sqrt{2\log 2}\right\} = \min\left\{O\left(\sqrt{\frac{\log|\mathscr{H}_d|}{n}}\right), O(1)\right\}. \tag{5}$$

We set the constant term to 0.5, because the Rademacher complexity of $\mathscr{L}$ is naturally bounded by 0.5 (see Fact C.1 in Appendix C).

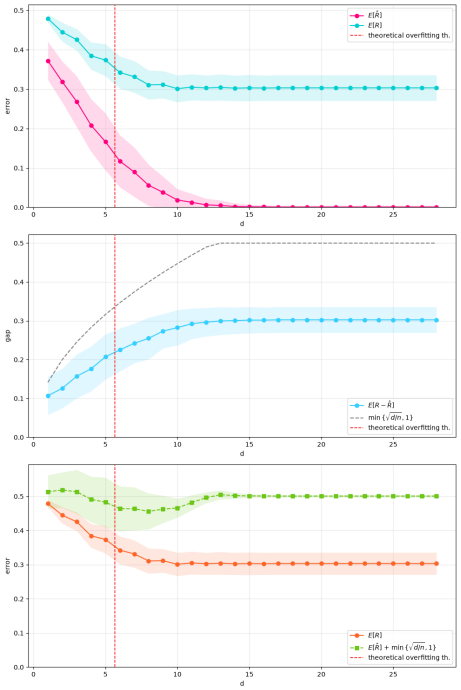


FIGURE 2. Rademacher and Massart bound (Theorem 3.8), same setting as Figure 1.

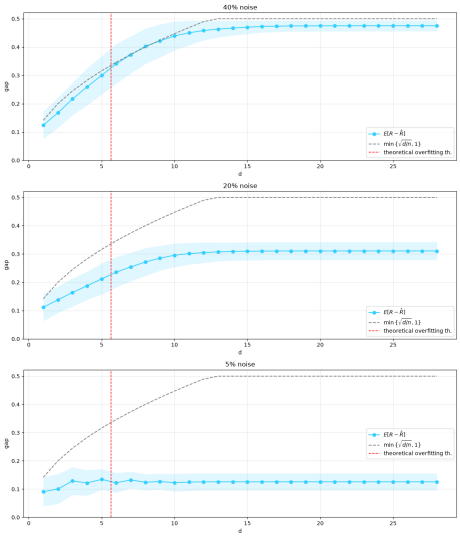


FIGURE 3. Rademacher and Massart bound against the gap for different noise levels.

We plot bound (5) against the empirical error curve in Figure 2. For smaller depths, the bound is the same as in Figure 1. From a depth of 13 onwards, the minimum is instead attained by the constant term, hence the bound saturates. This reflects

the fact that beyond a certain depth, the number of distinct classification vectors no longer increases with d through the mechanism explained above, hence the test error does not change.

At the same time, this mechanism is myopic to the other aspects of the structure of the learning problem. Indeed, Figure 3 shows how the gap curve is sensitive to different levels of noise in the dataset, as measured by the σ coefficient (standard deviation). We use $\sigma \in \{12.5, 3.5, 1\}$, which flip on average 40%, 20% and 5% of the labels. Specifically, it remains flatter with less noise, indicating both that the model performs better and that the difference between train and test error is lower (see Appendix B for experiments showing how the empirical curves change with noise). The bound curve, instead, remains unchanged, failing to capture that the noise present in the data has an effect on the generalization performance of the algorithm.

4. Dependence on the best risk

In this section, we implement a different approach and new theoretical tools to look for a noise-sensitive bound.

Let

$$N_S(\mathscr{H}) := |\{(\ell_h(x_1, y_1), \dots, \ell_h(x_n, y_n) : h \in \mathscr{H}, S = ((x_i, y_i))_{i=1}^n\}|$$

be a realization of $N_D(\mathscr{H})$ for a fixed dataset S, i.e. the number of all possible loss vectors for the hypotheses in $\mathscr{H}$ given S. Moreover, let

$$N_n(\mathscr{H}) := \sup_{S \in (\mathcal{X} \times \mathcal{Y})^n} N_S(\mathscr{H})$$

be the maximum possible cardinality of the set of loss vectors for the hypotheses in $\mathscr{H}$ over all datasets of size n. Let ℓ be the zero-one loss for the rest of this section.

Lemma 4.1 (Relative deviation)**.** *For $\varepsilon > 0$ with $n\varepsilon^2 \geq 8$,*

$$\Pr\left[\exists h \in \mathscr{H} : (R - \hat{R}_D)(h) > \varepsilon\sqrt{R(h)}\right] \leq 2N_{2n}(\mathscr{H})e^{-n\varepsilon^2/16}.$$

The result above is a concentration result for the random variable $\hat{R}_D(h)$ in terms of the true risk $R(h)$. To obtain it, the key ingredient is Chebyshev inequality (Theorem A.2) that bounds the deviation of a random variable in terms of its variance. When a random variable is bounded, moreover, its variance is controlled by its expected risk because, intuitively, it cannot fluctuate much around its center. In particular, for $\ell_h(X, Y)$ that is bounded and Bernoulli, we have

$$\mathrm{Var}(\ell_h) = \mathbb{E}[\ell_h](1 - \mathbb{E}[\ell_h]) \leq \mathbb{E}[\ell_h] = R(h).$$

This explains how we get a bound on the deviation relative to the true risk. The bound we obtain, moreover, depends on $N_{2n}(\mathscr{H})$, the maximum number of loss vectors for datasets of size $2n$. This quantity arises naturally because the result uses a symmetrization technique. Also, as presented in Section 3, this technique considers two independent datasets D and D' and we count loss vectors over their union, hence the size $2n$ instead of n. Note that this result can be obtained also in terms of $N_{D \cup D'}(\mathscr{H})$, but to ease computations and avoid having to account for the randomness of the data D we prefer taking the supremum.

In high probability, and isolating the empirical risk $\hat{R}_D$ on the right hand side, we obtain the following result.

Theorem 4.2 (Bound with empirical risk)**.** *With probability at least* $1-\delta$*,*

$$(R-\hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\frac{\hat{R}_D(\hat{h}_D)\log(2N_{2n}(\mathscr{H})/\delta)}{n}} + 16\frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n}$$

We now want to bound the empirical risk of the ERM $\hat{R}_D(\hat{h}_D)$ in the right-hand side of the result above.

Let

$$R^*_{\mathscr{H}} = \min_{h\in\mathscr{H}} R(h)$$

be the minimum risk attainable in the hypotheses set and let

$$h_b \in \arg\min_{h\in\mathscr{H}} R(h)$$

be the best hypothesis, i.e. the one attaining $R^*_{\mathscr{H}}$. Again, at least one such model exists since $\mathscr{H}$ is finite and nonempty.

We have

$$\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h_b) \tag{6}$$

surely, by definition of ERM. Moreover, Bernstein's inequality (Theorem A.3) applied to $\hat{R}_D(h_b)$ gives

$$\hat{R}_D(h_b) \leq R^*_{\mathscr{H}} + \sqrt{\frac{2R^*_{\mathscr{H}}\log(2/\delta)}{n}} + \frac{2\log(2/\delta)}{3n}. \tag{7}$$

Note that the dependence on the best risk $R^*_{\mathscr{H}}$ appears because Bernstein's is a concentration result in terms of the variance, like Chebyshev's. As explained above, the variance of a bounded random variable like ℓ_{h_b} is $\mathrm{Var}(\ell_{h_b}) \leq \mathbb{E}[\ell_{h_b}] = R^*_{\mathscr{H}}$.

Therefore, inequalities (6) and (7) allow to bound $\hat{R}(\hat{h}_D)$ in terms of the best risk. Substituting this into Theorem 4.2 yields a bound on the gap that depends on the best risk, as the following result shows.

Corollary 4.3 (Bound with best risk)**.** *With probability at least* $1-\delta$*,*

$$(R-\hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\frac{R^*_{\mathscr{H}}\log(4N_{2n}(\mathscr{H})/\delta)}{n}} + 20\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}$$

Finally, we make the dependence on the noise explicit.

Let

$$R^* := \inf_{h:\mathcal{X}\to\mathcal{Y}} R(h)$$

be the Bayes risk. Note that the infimum in the definition runs over a superset of $\mathscr{H}$, hence $R^* \leq R^*_{\mathscr{H}}$. The best risk can be decomposed as

$$R^*_{\mathscr{H}} = \underbrace{R^*}_{\text{Bayes risk}} + \underbrace{R^*_{\mathscr{H}} - R^*}_{\text{Approximation error}} =: R^* + A_{\mathscr{H}}$$

where

- the Bayes risk R^* measures the classification error due to the irreducible noise ε, as it is the minimal error that any function $h : \mathcal{X} \to \mathcal{Y}$ necessarily makes;
- the approximation error $A_{\mathscr{H}}$ is the excess error we make given that we consider a restricted set of predictors, $\mathscr{H}$.

Note that if there is no noise, then $R^* = 0$ and the Bayes risk is attained by the data-generating function h^*. Moreover, if the model attaining R^* lives in the hypotheses class, then $A_{\mathscr{H}} = 0$.

Substituting into Corollary 4.3 we obtain the following result.

Corollary 4.4. *With probability* $1 - \delta$*,*

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\frac{(R^* + A_{\mathscr{H}})\log(4N_{2n}(\mathscr{H})/\delta)}{n}} + 20\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}$$

Tail integrating and using the counting lemma (Lemma 3.7) we obtain the following result.

Theorem 4.5 (Noise and approximation error bound)**.** *Let* ℓ *be the zero-one loss. Then*

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \leq \min\left\{4\sqrt{\frac{(R^* + A_{\mathscr{H}})\left(\log(4|\mathscr{H}|) + 1\right)}{n}} + 20\,\frac{\log(4|\mathscr{H}|) + 1}{n},\right.$$
$$\left. 4\sqrt{\frac{(R^* + A_{\mathscr{H}})\left((2n+2)\log 2 + 1\right)}{n}} + 20\,\frac{(2n+2)\log 2 + 1}{n}\right\}$$

The result above is a bound on the expected gap at the ERM that explicitly depends on the noise R^* while taking into account the finiteness of our problem via the counting lemma. Next, we plot it and investigate how it relates to the empirical error curve.

Experimental design

As compared to Sections 2 and 3 that used a Gaussian noise, we switch to a label-flip noise with flip probability $1-p$. This makes the computation of the Bayes risk and of the best risk easier. Therefore, we generate the labels as

$$y \leftarrow \operatorname{sign}(h^*(x)) \cdot b$$

where h^* is the sine function (3) and b is a random variable taking value 1 with probability p and -1 with probability $1-p$. We use $p \in \{0.6, 0.8, 0.95\}$.

Since $p \geq 0.5$, the Bayes risk is

$$R^* = 1 - p.$$

This is because the hypotheses attaining the Bayes risk selects the most probable label for each input often (*Bayes classifier* in Section 2 of [4]). When $\Pr(\text{flip}) = 1-p < 0.5$, the most probable label is the noiseless one determined by $\operatorname{sign}(h^*(x))$, which gives error $1-p$. The best risk $R^*_{\mathscr{H}_d}$ is computed exactly for each d using dynamic programming (see Appendix D for details on the computation of the best risk).

The rest of the experimental design is as above.

As before, we ignore the constants and plot

$$\min\left\{O\left(\sqrt{\frac{(R^* + A_{\mathscr{H}_d})\log|\mathscr{H}_d|}{n}} + \frac{\log|\mathscr{H}_d|}{n}\right), O\left(\sqrt{R^* + A_{\mathscr{H}_d}} + 1\right)\right\} \tag{8}$$

We plot bound (8) in Figure 4. The shape of the bound curve is qualitatively the same as in Figure 2 from Section 3, with the bound generally increasing at smaller depths before reaching a plateau. Figure 5 allows to inspect the graph with the empirical gap and the bound for noise levels of $5\%, 20\%$, and 40%. We notice that both the empirical curve and the bound curve shift upwards as the noise level increases. Hence, the bound correctly captures the sensitivity of the empirical test

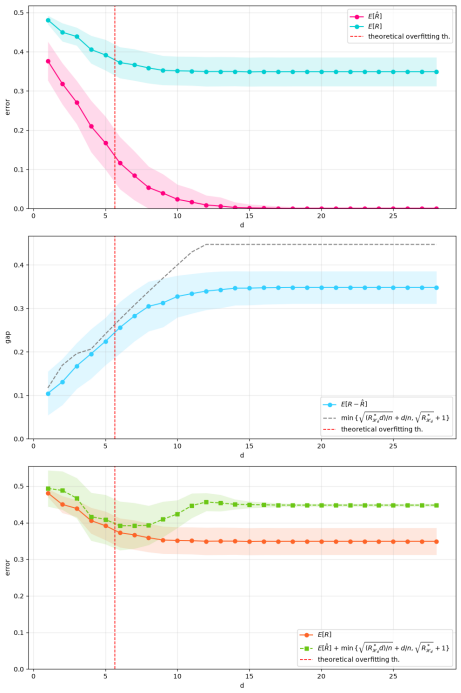


FIGURE 4. Noise and approximation error bound (Theorem 4.5), same setting as Figure 1 with label-flip noise at 20% ($p = 0.8$).

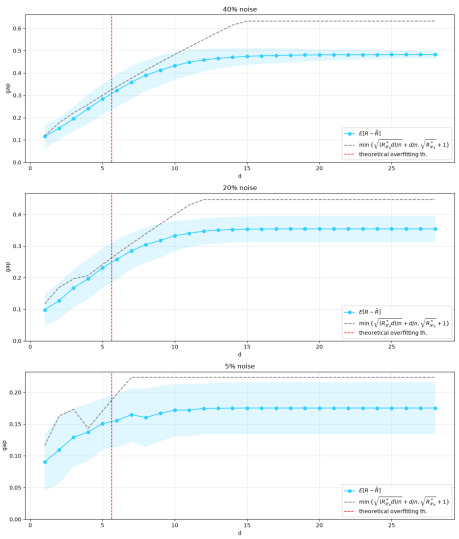


FIGURE 5. Noise and approximation error bound against the gap, for different noise levels.

error to different amounts of noise. Moreover, we notice that for smaller depths the bound presents a non-linear and non-monotonic part, which is more evident with noise at 5%. We now aim at obtaining a better understanding of this curve.

The bound changes with the noise because it is a function of the best risk $R^*_{\mathscr{H}}$. Figure 6 allows to investigate how $R^*_{\mathscr{H}}$ is affected by different noise levels (blue line). Let d_c be the depth at which the best risk saturates, which is the same in the three graphs. We make two observations:

(1) for $d < d_c$, the best risk decreases faster with less noise;

(2) for $d \geq d_c$, the best risk saturates and the saturating level is lower with less noise.

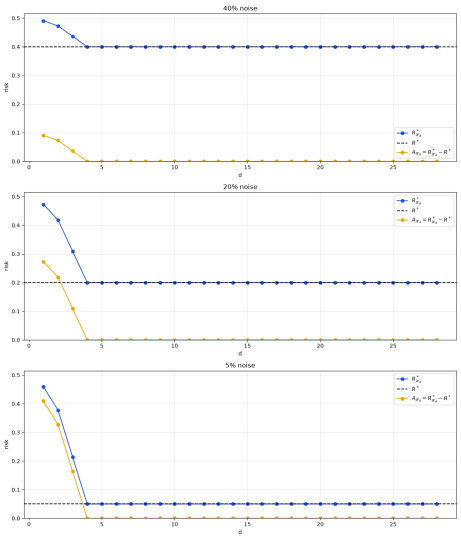


FIGURE 6. Best risk $R^*_{\mathscr{H}_d}$ split into Bayes risk R^* and approximation error $A_{\mathscr{H}_d}$, for different noise levels.

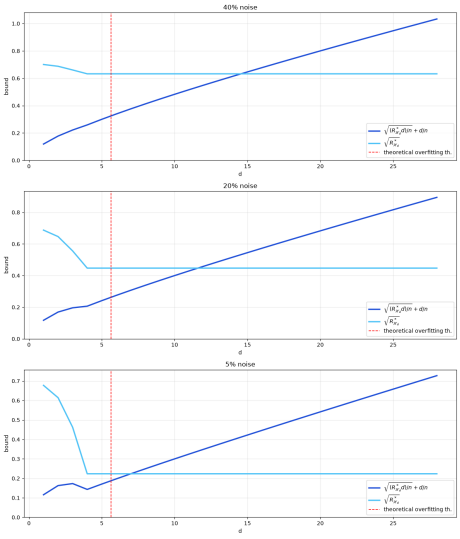


FIGURE 7. The two terms in the minimum of the noise and approximation error bound, for different noise levels.

We suppose that the first dynamic is due to the fact that, with less noise, there are fewer contiguous block with the same label, therefore a marginal increase in the depth from d to d' (i.e., allowing $2^{d'} - 2^d$ additional leaves) results in a larger decrease of the error, because, morally, more inputs will change classification in a single shot. The second observation regarding the saturating level, instead, is directly determined by the irreducible noise in the data. In fact, the best risk saturates exactly at the Bayes risk (black, dashed line). Through these two mechanisms, the bound, seen as a function of the best risk, captures the dependence of the test error of the ERM on the noise. In particular, this explains the shape of the bound that we observed in Figure 5. The initial non-linearity is explained by mechanism (1), while the upward shift of the bound curve as noise increases is motivated by (2).

More in detail, we investigate how $R^*_{\mathscr{H}}$ enters the bounding function, determining its specific shape. This can be inspected in Figure 7, which shows the two terms in the minimum defining the bound. Note that both terms depend on $R^*_{\mathscr{H}}$, hence

they both change with the noise. For smaller depths, the lowest curve is the blue line, and the non-linearity is determined by the interaction between the best risk (decreasing) and the depth (increasing), via $\sqrt{R^*_{\mathscr{H}} d}$. Since with less noise the best risk decreases faster, this non-linearity is more evident in this case. Then, once $d \geq d_c$, the curve increases linearly because $R^*_{\mathscr{H}}$ becomes constant. For larger depths, the curve attaining the minimum is the light-blue curve, which is constant at the Bayes risk $\sqrt{R^*}$.

5. Growth function

In this section, we introduce the growth function, a new measure of complexity of the hypotheses class that considers classification vectors instead of loss vectors. This tool allows to make the intuition on the saturation of the test error due to characteristics of the decision tree algorithm that we discussed in Section 3 more explicit.

Definition 5.1 (Growth function of $\mathscr{H}$)**.** *The growth function of $\mathscr{H}$ is*

$$\Pi_n(\mathscr{H}) := \sup_{x_1,\dots,x_n \in \mathcal{X}} \left|\{(h(x_1),\dots,h(x_n)) : h \in \mathscr{H}\}\right|.$$

The following extends Lemma 3.7.

Lemma 5.2 (Counting with growth function)**.** *Let ℓ be the zero-one loss. Then*

$$1 \leq N_D(\mathscr{H}) \leq \Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$$

surely. Moreover,

$$\Pi_n(\mathscr{H}) = N_n(\mathscr{H}) = \sup_{S \in (\mathcal{X}\times\mathcal{Y})^n} N_S(\mathscr{H}).$$

Morally, applying ℓ to the image set of h can at most decrease the number of vectors in the set, hence $N_D \leq N_n = \Pi_n$. At the same time, the images of the functions in $\mathscr{H}$ are at most as many as the functions in $\mathscr{H}$, and always less than or equal to 2^n, the number of all possible n-long sequences of 1 and -1.

The relation between N_D and Π_n makes it immediate to obtain growth-function versions of the bounds in Sections 3 and 4 (substitute $N(D) \leq \Pi_n$ into Lemma 3.6 and $N_{2n} = \Pi_{2n}$ into Theorem 4.2). Substituting also $\Pi_n(\mathscr{H}) \leq \min\{|\mathscr{H}|, 2^n\}$ yields exactly bounds (5) and (8). The bounds are the same, but the intuition that they saturate because the number of distinct classification vectors is at most 2^n is made more explicit. In other words, when we evaluate the performance of an algorithm, the models in $\mathscr{H}$ that the algorithm can learn are equivalent if they give the same classification. This result takes this into account when measuring the richness of the hypotheses class $\mathscr{H}$.

Therefore, the following Theorem constitutes our final result.

Theorem 5.3 (Bound with growth function)**.** *Let ℓ be the zero-one loss. Then*

$$\mathbb{E}_D\left[(R-\hat{R}_D)(\hat{h}_D)\right] \leq 4\sqrt{\frac{(R^*+A_{\mathscr{H}})\left(\log(4\Pi_{2n}(\mathscr{H}))+1\right)}{n}}+20\,\frac{\log(4\Pi_{2n}(\mathscr{H}))+1}{n}$$

with

$$\Pi_{2n}(\mathscr{H}) \leq \min\left\{|\mathscr{H}|, 2^{2n}\right\}.$$

This result provides an intuition on the drivers of the test error of decision trees. One one side, the bound it provides depends on the maximum number of classification vectors Π_{2n}, representing the maximum cardinality of the hypotheses class in practice. On the other side, the bound is a function of the noise R^* and the approximation error $A_{\mathscr{H}}$, capturing the effect that the noise in the data and our choice of hypotheses have on generalization. We empirically validate this with the experiments presented in Figure 4 to 7. As such, the result provides a generalization bound that captures the behavior of the test error of decision trees as model complexity increases.

6. Conclusions

This work has inquired and empirically validated classical generalization bounds for decision trees. We have started from a bound using Hoeffing's inequality and a union bound over the hypotehses set, which did not saturate. Then, we have taken into account the fact that the class of functions that decision trees can learn is less complex than the one considered initially, due to the finite nature of the algorithm. By including this in our analysis via Lemma 5.2, we have obtained a saturating bound. Finally, we have noticed that the generalization performance of decision trees is affected by the level of noise in the data. Thus, we have obtained a bound that is a function of the lowest risk that a decision tree can attain, that depends on the noise through the Bayes risk. This, combined with the result on the actual complexity of the hypotheses space, has given a bound that saturates and that is sensitive to the noise. Theorem 5.3 summarizes this result.

The main limitation of this work is given by the lack of an analysis of the difference between the selected model that we consider for theortical results, i.e. the Empirical Risk Minimizer, and the model learned in practie by the DecisionTreeClassifier algorithm. In fact, this is a greedy algorithm that not always selected the best posssible model.

Apart from analyzing this criticality, the most promising direction for extending this thesis in the future is to apply the framework and approach proposed here to other algorithms. Specifically, we propose Random Fourier Features, Random Forests, and neural networks trained with Stochastic Gradient Descent as main developments.

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Appendix A. Proofs

We first recall some basic results that are used throughout, and then give the proofs of the results in the main text.

Theorem A.1 (Hoeffding's inequality)**.** *Let $Z_1, \dots, Z_n$ be independent real-valued random variables with $Z_i \in [a_i, b_i]$ almost surely for some $a_i, b_i \in \mathbb{R}$, and let $\mu = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[Z_i]$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^n Z_i - \mu \geq \varepsilon\right] \leq \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Theorem A.2 (Chebyshev's inequality)**.** *Let Z be a real-valued random variable with finite variance. Then, for every $\varepsilon > 0$,*

$$\Pr\left[|Z - \mathbb{E}[Z]| \geq \varepsilon\right] \leq \frac{\operatorname{Var}(Z)}{\varepsilon^2}.$$

Theorem A.3 (Bernstein's inequality)**.** *Let $Z_1, \dots, Z_n$ be independent real-valued random variables with $\mathbb{E}[Z_i] = 0$ and $Z_i \leq M$ almost surely for some $M > 0$, and let $v = \frac{1}{n}\sum_{i=1}^n \operatorname{Var}(Z_i)$. Then, for every $\varepsilon > 0$,*

$$\Pr\left[\frac{1}{n}\sum_{i=1}^n Z_i \geq \varepsilon\right] \leq \exp\left(-\frac{n\varepsilon^2}{2(v + M\varepsilon/3)}\right).$$

Proposition A.4 (Union bound)**.** *For any finite or countable collection of events $A_1, A_2, \dots$,*

$$\Pr\left[\bigcup_i A_i\right] \leq \sum_i \Pr[A_i].$$

Theorem A.5 (Hoeffding's lemma)**.** *Let Z be a real-valued random variable with $\mathbb{E}[Z] = 0$ and $|Z| \leq a$ almost surely for some $a \in \mathbb{R}$. Then, for every $t \in \mathbb{R}$,*

$$\mathbb{E}\left[e^{tZ}\right] \leq e^{t^2 a^2/2}.$$

Proposition A.6 (Tail integration)**.** *For any non-negative random variable Z,*

$$\mathbb{E}[Z] = \int_0^\infty \Pr[Z \geq \varepsilon]\ \mathrm{d}\varepsilon.$$

Proposition A.7 (Gaussian tail bound)**.** *For every $c > 0$ and $x_0 > 0$,*

$$\int_{x_0}^\infty e^{-cx^2}\ \mathrm{d}x \leq \frac{1}{2cx_0} e^{-cx_0^2}.$$

Proof of Proposition 2.3. Let $\ell : A \to \mathbb{R}^+$ satisfy Assumption 2.2 with constants C and C'. Then

$$\begin{aligned}\ell(a) &= |\ell(a)| \\ &= |\ell(a) - \ell(0) + \ell(0)| \\ &\leq |\ell(a) - \ell(0)| + |\ell(0)| \\ &\leq C|a| + C' \\ &\leq CB + C' =: M\end{aligned}$$

□

Proof of Theorem 2.6. We prove the lower bound first. Since $\hat{h}_D \in \arg\min_{h \in \mathscr{H}} \hat{R}_D(h)$, we have

$$\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h) \qquad \forall h \in \mathscr{H},$$

and taking expectations

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq \mathbb{E}_D[\hat{R}_D(h)] \qquad \forall h \in \mathscr{H}.$$

For a fixed $h \in \mathscr{H}$, which does not depend on D, the empirical risk is an unbiased estimate of the true risk, by linearity and because the (X_i, Y_i) are distributed as (X, Y),

$$\mathbb{E}_D[\hat{R}_D(h)] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\ell(h(X_i), Y_i)\right] = R(h).$$

Hence

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq R(h) \qquad \forall h \in \mathscr{H}.$$

Now take S a realization of D and consider in particular $h = \hat{h}_S \in \mathscr{H}$,

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq R(\hat{h}_S).$$

Since this holds for every realization S of D, we have $\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq R(\hat{h}_D)$ almost surely and, taking expectations,

$$\mathbb{E}_D[\hat{R}_D(\hat{h}_D)] \leq \mathbb{E}_D[R(\hat{h}_D)].$$

Next, we prove the upper bound. Fix $h \in \mathscr{H}$ and let $Z_i \coloneqq -\ell(h(X_i), Y_i)$, $i = 1, \dots, n$, so that

$$\frac{1}{n}\sum_{i=1}^{n} Z_i - \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Z_i] = (R - \hat{R}_D)(h).$$

We want to apply Hoeffding's inequality (Theorem A.1). The Z_i are independent, since the (X_i, Y_i) are, and $Z_i \in [-M, 0]$ by non-negativity of the loss function and Proposition 2.3. Hence, with $a_i = -M$ and $b_i = 0$, we obtain

$$\Pr\left[(R - \hat{R}_D)(h) \geq \varepsilon\right] \leq e^{-2n\varepsilon^2/M^2}. \tag{9}$$

Now, to obtain a bound at $\hat{h}_D$, which depends on D, notice that if $(R - \hat{R}_D)(\hat{h}_D) \geq \varepsilon$ then $(R - \hat{R}_D)(h) \geq \varepsilon$ for some $h \in \mathscr{H}$. Therefore, by the union bound (Proposition A.4) and (9),

$$\begin{aligned}
\Pr\left[(R - \hat{R}_D)(\hat{h}_D) \geq \varepsilon\right] &\leq \Pr\left[\bigcup_{h \in \mathscr{H}} \left\{(R - \hat{R}_D)(h) \geq \varepsilon\right\}\right] \\
&\leq \sum_{h \in \mathscr{H}} \Pr\left[(R - \hat{R}_D)(h) \geq \varepsilon\right] \\
&\leq |\mathscr{H}| e^{-2n\varepsilon^2/M^2}.
\end{aligned} \tag{10}$$

To obtain a bound in expected value from (10), we tail integrate. Let $X \coloneqq (R - \hat{R}_D)(\hat{h}_D)$ and $X^+ \coloneqq \max\{X, 0\}$, which is non-negative, so that by Proposition A.6

$$\mathbb{E}_D[X] \leq \mathbb{E}_D[X^+] = \int_0^\infty \Pr[X^+ \geq \varepsilon] \ \mathrm{d}\varepsilon = \int_0^\infty \Pr[X \geq \varepsilon] \ \mathrm{d}\varepsilon.$$

Let $\alpha \coloneqq |\mathscr{H}|$ and $c \coloneqq 2n/M^2$, so that (10) reads $\Pr[X \geq \varepsilon] \leq \alpha e^{-c\varepsilon^2}$. This bound is good for large values of ε, but for small ε it exceeds 1, hence we use instead

$$\Pr[X \geq \varepsilon] \leq \min\{1, \alpha e^{-c\varepsilon^2}\}.$$

We split the integral at the threshold ε_0 where $\alpha e^{-c\varepsilon_0^2} = 1$, i.e. $\varepsilon_0 = \sqrt{\log \alpha / c}$. Using the Gaussian tail bound (Proposition A.7) for the second integral, we obtain

$$\mathbb{E}_D[X] \leq \int_0^{\varepsilon_0} 1 \ \mathrm{d}\varepsilon + \int_{\varepsilon_0}^\infty \alpha e^{-c\varepsilon^2} \ \mathrm{d}\varepsilon \leq \varepsilon_0 + \frac{\alpha}{2c\varepsilon_0} e^{-c\varepsilon_0^2} = \sqrt{\frac{\log \alpha}{c}} + \frac{1}{2\sqrt{c \log \alpha}},$$

where in the last equality we used $\alpha e^{-c\varepsilon_0^2} = 1$. Finally, assuming $\alpha \geq e^{1/2}$, which holds as soon as $|\mathscr{H}| \geq 2$, we have $\frac{1}{2\sqrt{c \log \alpha}} \leq \sqrt{\frac{\log \alpha}{c}}$, and hence

$$\mathbb{E}_D[X] \leq 2\sqrt{\frac{\log \alpha}{c}} = 2M\sqrt{\frac{\log |\mathscr{H}|}{2n}}.$$

□

Proof of Theorem 3.3. Let $Z \coloneqq (X, Y)$ and $Z_i \coloneqq (X_i, Y_i)$, so that

$$R(h) = \mathbb{E}[\ell_h(Z)] \qquad \text{and} \qquad \hat{R}_D(h) = \frac{1}{n}\sum_{i=1}^{n} \ell_h(Z_i).$$

Introduce an independent copy $D' = (Z'_1, \dots, Z'_n)$ of the dataset D. Since the Z'_i are distributed as Z and are independent of D, we rewrite

$$R(h) = \mathbb{E}_{D'}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i)\right] = \mathbb{E}_{D'}\left[\frac{1}{n}\sum_{i=1}^{n} \ell_h(Z'_i) \,\middle|\, D\right].$$

Then,

$$\mathbb{E}_D\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] = \mathbb{E}_D\left[\sup_{\ell_h \in \mathscr{L}} \mathbb{E}_{D'}\left[\frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right) \,\middle|\, D\right]\right].$$

since, again, Z_i is independent of D' hence it can enter the expectation. Since the supremum of expectations is at most the expectation of the supremum, and by the law of total expectation, we obtain

$$\mathbb{E}_D\left[\sup_{\ell_h \in \mathscr{L}} (R - \hat{R}_D)(h)\right] \leq \mathbb{E}_{D,D'}\left[\sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right)\right]. \tag{11}$$

Now we apply the symmetrization trick. Let $\mathscr{H} = \{h_1, \dots, h_m\}$, with $m = |\mathscr{H}|$. Since Z_i and Z'_i are i.i.d. for each $i = 1, \dots, n$, swapping them does not change the joint distribution, i.e.

$$\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \overset{d}{=} \left(\ell_{h_j}(Z_i) - \ell_{h_j}(Z'_i)\right)_{j=1}^{m}.$$

This means that, for any fixed $(e_1, \dots, e_n) \in \{-1, 1\}^n$,

$$\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)_{j=1}^{m} \overset{d}{=} \left(e_i\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m}.$$

The pairs $(Z_1, Z'_1), \dots, (Z_n, Z'_n)$ are also independent, therefore equality in distribution holds also when taking the sum, i.e.

$$\left(\frac{1}{n}\sum_{i=1}^{n}\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m} \overset{d}{=} \left(\frac{1}{n}\sum_{i=1}^{n} e_i\left(\ell_{h_j}(Z'_i) - \ell_{h_j}(Z_i)\right)\right)_{j=1}^{m}.$$

Finally, because the m random variables are jointly equally distributed, we have equality in distribution of the suprema

$$\sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} \left(\ell_h(Z'_i) - \ell_h(Z_i)\right) \overset{d}{=} \sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} e_i\left(\ell_h(Z'_i) - \ell_h(Z_i)\right).$$

Because the two sides in the previous equation have the same distribution, their expected value is the same

$$\mathbb{E}_{D,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right] = \mathbb{E}_{D,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}e_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right].$$

Note that until now we have considered only the disorder of the data D, D'. Because this holds for any value of $(e_1,\dots,e_n)\in\{-1,1\}^n$, it holds surely for the random variable $\xi=\xi_1,\dots,\xi_n$, i.e.

$$\mathbb{E}_{D,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right] = \mathbb{E}_{D,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right]$$

surely. Taking the expected value over the disorder of ξ, we obtain

$$\mathbb{E}_{D,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right] = \mathbb{E}_{\xi,D,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right]$$

where the left hand side is independent of ξ. Plugging the result into (11), we have

$$\mathbb{E}_{D}\left[\sup_{\ell_h\in\mathscr{L}}(R-\hat{R}_D)(h)\right] \le \mathbb{E}_{\xi,D,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right)\right]$$

Now we split the sum over $i=1,\dots,n$ into

$$\frac{1}{n}\sum_{i=1}^{n}\xi_i\left(\ell_h(Z_i')-\ell_h(Z_i)\right) = \frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i') + \frac{1}{n}\sum_{i=1}^{n}\left(-\xi_i\ell_h(Z_i)\right)$$

Then, we use that the supremum of the sum is bounded by the sum of the suprema,

$$\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')+\frac{1}{n}\sum_{i=1}^{n}\left(-\xi_i\ell_h(Z_i)\right) \le \sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')+\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\left(-\xi_i\ell_h(Z_i)\right)$$

and that the expectation of the sum is the sum of the expectations, obtaining

$$\mathbb{E}_{D}\left[\sup_{\ell_h\in\mathscr{L}}(R-\hat{R}_D)(h)\right] \le \mathbb{E}_{\xi,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')\right]+\mathbb{E}_{\xi,D}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}(-\xi_i)\ell_h(Z_i)\right]$$

We also observe that we can replace $-\xi_i$ with ξ_i, because they have the same distributions, hence the same expected value, and are independent of the datasets. Therefore,

$$\begin{aligned}\mathbb{E}_{D}\left[\sup_{\ell_h\in\mathscr{L}}(R-\hat{R}_D)(h)\right] &\le \mathbb{E}_{\xi,D'}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i')\right]+\mathbb{E}_{\xi,D}\left[\sup_{\ell_h\in\mathscr{L}}\frac{1}{n}\sum_{i=1}^{n}\xi_i\ell_h(Z_i)\right]\\ &= 2\mathcal{R}_n(\mathscr{L}).\end{aligned}$$

□

Proof of Lemma 3.4. For any $t > 0$,

$$\begin{aligned}
\exp\left(t\,\mathbb{E}_{\xi,D}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) &\le \mathbb{E}_{\xi,D}\left[\exp\left(t\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&= \mathbb{E}_{\xi,D}\left[\sup_{\ell_h\in\mathscr{L}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&\le \mathbb{E}_{\xi,D}\left[\sum_{\ell_h\in\mathscr{L}}\exp\left(t\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right)\right] \\
&= \mathbb{E}_{D}\left[\sum_{\ell_h\in\mathscr{L}}\prod_{i=1}^{n}\mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\right]\right],
\end{aligned} \tag{12}$$

where we use Jensen's inequality for the convex function $\exp(\cdot)$ in the first inequality, that exp is increasing in the first equality, we bound the supremum with the sum (all terms are positive) in the second inequality, and we use the independence of the ξ_i's, and their independence of D, in the last equality.

Now we apply Hoeffding's lemma (Theorem A.5) to each factor for fixed $(X_i, Y_i) = (x_i, y_i)$ taking any value in $\mathcal{X}\times\mathcal{Y}$. The random variable $\xi_i\ell_h(x_i,y_i)$ is centered, since $\mathbb{E}[\xi_i] = 0$, and it satisfies $|\xi_i\ell_h(x_i,y_i)| = \ell_h(x_i,y_i) \le M$ by non-negativity of the loss and Proposition 2.3, hence it satisfies the assumptions of Hoeffding's lemma. We obtain

$$\mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i\ell_h(x_i,y_i)\right)\right] \le e^{t^2M^2/2}.$$

Because this holds for any $(x_i, y_i) \in \mathcal{X}\times\mathcal{Y}$, we have

$$\mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\right] \le e^{t^2M^2/2}$$

surely. Plugging this into (12),

$$\exp\left(t\,\mathbb{E}_{\xi,D}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) \le \mathbb{E}_{D}\left[\sum_{\ell_h\in\mathscr{L}}\prod_{i=1}^{n}e^{t^2M^2/2}\right] = |\mathscr{L}|\,e^{nt^2M^2/2}.$$

Taking the logarithm of both sides,

$$t\,\mathbb{E}_{\xi,D}\left[\sup_{\ell_h\in\mathscr{L}}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right] \le \log|\mathscr{L}| + \frac{nt^2M^2}{2},$$

and dividing by $tn > 0$,

$$\mathcal{R}_n(\mathscr{L}) \le \frac{\log|\mathscr{L}|}{tn} + \frac{tM^2}{2} =: b(t) \tag{13}$$

Since (13) holds for every $t > 0$, we look for the tightest bound. We set the derivative in t to zero

$$\frac{\mathrm{d}b(t)}{\mathrm{d}t} = -\frac{\log|\mathscr{L}|}{t^2 n} + \frac{M^2}{2} = 0 \quad \Longrightarrow \quad t^\star = \sqrt{\frac{2\log|\mathscr{L}|}{nM^2}}.$$

Observe that since it is fair to assume $|\mathscr{L}| > 1$, the second derivative is nonnegative for any t

$$\frac{2\log|\mathscr{L}|}{t^3 n} \geq 0 \quad \forall t$$

hence $t^\star$ is a minimizer (if $|\mathscr{L}| = 1$, the bound follows by letting $t \to \infty$). Plugging $t^\star$ into (13) yields

$$\mathcal{R}_n(\mathscr{L}) \leq M\sqrt{\frac{2\log|\mathscr{L}|}{n}}$$

□

Proof of Lemma 3.6. Most of the proof is similar to Proof of Lemma 3.4.

Let $(\ell_h(X_1, Y_1), \ldots, \ell_h(X_n, Y_n)) = \ell(D)$. For any $t > 0$,

$$\exp\left(t\,\mathbb{E}_{\xi,D}\left[\sup_{\ell(D)\in\mathscr{L}(D)}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) \leq \mathbb{E}_D\left[\sum_{\ell(D)\in\mathscr{L}(D)}\prod_{i=1}^{n}\mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\right]\right]$$

following the same steps as in Proof of Lemma 3.4, all of which are independent of the data disorder, as they can be made fixing an observation S.

Now we apply Hoeffding's lemma (Theorem A.5). These steps also regard the argument of the expectation over the Rademacher variables disorder. We obtain

$$\mathbb{E}_{\xi_i}\left[\exp\left(t\xi_i\ell_h(X_i,Y_i)\right)\right] \leq e^{t^2M^2/2}.$$

surely as in Proof of Lemma 3.4. Plugging this into (12),

$$\begin{aligned}\exp\left(t\,\mathbb{E}_{\xi,D}\left[\sup_{\ell(D)\in\mathscr{L}(D)}\sum_{i=1}^{n}\xi_i\ell_h(X_i,Y_i)\right]\right) &\leq \mathbb{E}_D\left[\sum_{\ell(D)\in\mathscr{L}(D)}\prod_{i=1}^{n}e^{t^2M^2/2}\right]\\ &= \mathbb{E}_D\left[N_D(\mathscr{H})e^{nt^2M^2/2}\right]\\ &= \mathbb{E}_D\left[N_D(\mathscr{H})\right]e^{nt^2M^2/2}.\end{aligned}$$

The rest of the proof follows exacltly as Proof of Lemma 3.4, replacing $|\mathscr{L}|$ with $\mathbb{E}_D\left[N_D(\mathscr{H})\right]$. □

Proof of Lemma 3.7. We prove that $N_S(\mathscr{H}) \le |\mathscr{H}|$ and $N_S(\mathscr{H}) \le 2^n$ for every $S \in (\mathcal{X} \times \mathcal{Y})^n$. Fix $S = (z_1, \dots, z_n)$ and consider the map

$$\Phi \colon \mathscr{H} \to \{0,1\}^n, \qquad h \mapsto \big(\ell_h(z_1), \dots, \ell_h(z_n)\big),$$

so that $N_S(\mathscr{H}) = |\Phi(\mathscr{H})|$. Since the image of a set under any map has cardinality at most that of the domain, $|\Phi(\mathscr{H})| \le |\mathscr{H}|$. Since $\Phi(\mathscr{H}) \subseteq \{0,1\}^n$, we also have $|\Phi(\mathscr{H})| \le |\{0,1\}^n| = 2^n$. Because this holds for every realization S of D, the claim for $N_D(\mathscr{H})$ follows. □

Proof of Theorem 3.8. Since $\hat{h}_S \in \mathscr{H}$ for any realization S of D,

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le \mathbb{E}_D\left[\sup_{h \in \mathscr{H}} (R - \hat{R}_D)(h)\right]$$

The proof is concluded applying Theorem 3.3, bounding the Rademacher complexity with Lemma 3.6, and bounding $N_D(\mathscr{H})$ with Lemma 3.7. □

Proof of Lemma 4.1. Recall $\mathscr{L} = \{(x,y) \mapsto \ell(y, h(x)) : h \in \mathscr{H}\}$, where ℓ is the zero-one loss. Applying the first inequality of Theorem 5.1 of [4], with $\mathbb{S}_{\mathcal{F}}(X_1^{2n}) = N_{2n}(\mathscr{H})$, we obtain that with probability at least $1 - \delta$, simultaneously for all $h \in \mathscr{H}$,

$$R(h) - \widehat{R}_D(h) \le 2\sqrt{\frac{R(h)\log(4N_{2n}(\mathscr{H})/\delta)}{n}}.$$

Set

$$\delta_\varepsilon = 4N_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon < 1$, applying the preceding inequality with $\delta = \delta_\varepsilon$ gives

$$\Pr\left[\exists h \in \mathscr{H} : R(h) - \widehat{R}_D(h) > \varepsilon\sqrt{R(h)}\right] \le 4N_{2n}(\mathscr{H})e^{-n\varepsilon^2/4}.$$

If $\delta_\varepsilon \ge 1$, this probability bound holds trivially. Finally, since $n\varepsilon^2 \ge 8$,

$$4e^{-n\varepsilon^2/4} \le 2e^{-n\varepsilon^2/16},$$

which proves the claim. □

Proof of Theorem 4.2. Let $\delta \in (0,1)$ and set

$$u \coloneqq \frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n} \qquad \text{and} \qquad \varepsilon \coloneqq 4\sqrt{u},$$

so that $2N_{2n}(\mathscr{H})e^{-n\varepsilon^2/16} = \delta$. Since $N_{2n}(\mathscr{H}) \geq 1$ and $\delta < 1$, we have $n\varepsilon^2 = 16\log(2N_{2n}(\mathscr{H})/\delta) \geq 16\log 2 \geq 8$, hence Lemma 4.1 applies and gives

$$\Pr\left[\exists h \in \mathscr{H} : (R - \hat{R}_D)(h) > \varepsilon\sqrt{R(h)}\right] \leq \delta.$$

Therefore, with probability at least $1-\delta$, we have $(R - \hat{R}_D)(h) \leq \varepsilon\sqrt{R(h)}$ for all $h \in \mathscr{H}$, and in particular for $h = \hat{h}_D \in \mathscr{H}$.

On this event, we isolate $\hat{R}_D(\hat{h}_D)$. Let $r \coloneqq R(\hat{h}_D)$ and $\hat{r} \coloneqq \hat{R}_D(\hat{h}_D)$, so that $r - \varepsilon\sqrt{r} - \hat{r} \leq 0$. This is a quadratic inequality in $x \coloneqq \sqrt{r} \geq 0$, namely $x^2 - \varepsilon x - \hat{r} \leq 0$, which implies that x is at most the largest root,

$$\sqrt{r} \leq \frac{\varepsilon + \sqrt{\varepsilon^2 + 4\hat{r}}}{2}.$$

Squaring both sides,

$$r \leq \frac{2\varepsilon^2 + 4\hat{r} + 2\varepsilon\sqrt{\varepsilon^2 + 4\hat{r}}}{4} = \hat{r} + \frac{\varepsilon^2}{2} + \frac{\varepsilon}{2}\sqrt{\varepsilon^2 + 4\hat{r}} \leq \hat{r} + \varepsilon^2 + \varepsilon\sqrt{\hat{r}},$$

where in the last inequality we used $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$. Hence, substituting $\varepsilon = 4\sqrt{u}$, with probability at least $1-\delta$,

$$\begin{aligned}(R - \hat{R}_D)(\hat{h}_D) = r - \hat{r} &\leq \varepsilon\sqrt{\hat{r}} + \varepsilon^2 \\ &= 4\sqrt{\frac{\hat{R}_D(\hat{h}_D)\log(2N_{2n}(\mathscr{H})/\delta)}{n}} + 16\frac{\log(2N_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

Proof of Corollary 4.3. Recall $h_0 = \arg\min_{h\in\mathscr{H}} R(h)$, $R(h_0) = R^*_{\mathscr{H}}$ by definition of $R^*_{\mathscr{H}}$ and $\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h_0)$ by definition of the ERM $\hat{h}_D$. Since h_0 is independent of the data, the variables $\ell(h_0(X_i), Y_i)$ are i.i.d. Bernoulli with mean $R^*_{\mathscr{H}}$ and variance $R^*_{\mathscr{H}}(1 - R^*_{\mathscr{H}}) \leq R^*_{\mathscr{H}}$. Bernstein inequality (Theorem A.3) with $M = 1$ gives, with probability at least $1 - \delta/2$,

$$\hat{R}_D(h_0) \leq R^*_{\mathscr{H}} + \sqrt{\frac{2R^*_{\mathscr{H}}\log(2/\delta)}{n}} + \frac{2\log(2/\delta)}{3n}.$$

Let $u \coloneqq \log(4N_{2n}(\mathscr{H})/\delta)/n$. Since $N_{2n}(\mathscr{H}) \geq 1$, $\log(2/\delta)/n \leq u$. Thus

$$\hat{R}_D(\hat{h}_D) \leq \hat{R}_D(h_0) \leq R^*_{\mathscr{H}} + 2\sqrt{R^*_{\mathscr{H}}u} + u = \left(\sqrt{R^*_{\mathscr{H}}} + \sqrt{u}\right)^2. \tag{14}$$

Theorem 4.2 at confidence $1 - \delta/2$ gives, with probability at least $1 - \delta/2$,

$$(R - \hat{R}_D)(\hat{h}_D) \leq 4\sqrt{\hat{R}_D(\hat{h}_D)\,u} + 16u. \tag{15}$$

By a union bound, both events hold simultaneously with probability at least $1-\delta$. Hence, plugging (14) in the RHS of (15), we obtain that with probability at least $1-\delta$,

$$\begin{aligned}(R-\hat{R}_D)(\hat{h}_D) &\le 4\sqrt{u}\left(\sqrt{R^*_{\mathscr{H}}}+\sqrt{u}\right)+16u \\ &= 4\sqrt{\frac{R^*_{\mathscr{H}}\log(4N_{2n}(\mathscr{H})/\delta)}{n}}+20\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}.\end{aligned}$$

□

Proof of Corollary 4.4. Substitute $R^*_{\mathscr{H}} = \sigma^2 + A_{\mathscr{H}}$ into Corollary 4.3. □

Proof of Theorem 4.5. Let $\text{gap} = (R-\hat{R}_D)(\hat{h}_D)$ and $C = \sqrt{\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n}}$. Corollary 4.3 gives

$$\begin{aligned}&\Pr\left[\text{gap} \le 4\sqrt{R^*_{\mathscr{H}}}\,C + 20C^2\right] \ge 1-\delta \\ \implies &\Pr\left[\text{gap} > 4\sqrt{R^*_{\mathscr{H}}}\,C + 20C^2\right] \le \delta\end{aligned}$$

Set $\varepsilon = 4\sqrt{R^*_{\mathscr{H}}}\,C + 20C^2$. Let us now find $\delta(\varepsilon)$:

$$20C^2+4\sqrt{R^*_{\mathscr{H}}}\,C-\varepsilon = 0 \implies C = \frac{-4\sqrt{R^*_{\mathscr{H}}}+\sqrt{16R^*_{\mathscr{H}}+80\varepsilon}}{40} = \frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}$$

So we have

$$\begin{aligned}\frac{\log(4N_{2n}(\mathscr{H})/\delta)}{n} &= \left(\frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}\right)^2 \\ \frac{4N_{2n}(\mathscr{H})}{\delta} &= \exp\left(n\left(\frac{\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}}{10}\right)^2\right)\end{aligned}$$

and hence

$$\delta(\varepsilon) = 4N_{2n}(\mathscr{H})\exp\left(-\frac{n}{100}\left(\sqrt{R^*_{\mathscr{H}}+5\varepsilon}-\sqrt{R^*_{\mathscr{H}}}\right)^2\right).$$

Using that $\mathbb{E}[X] = \int_0^\infty \Pr[X \ge \varepsilon]\,d\varepsilon$ for $X \ge 0$, applied to $\text{gap}^+ \coloneqq \max\{\text{gap}, 0\}$,

$$\mathbb{E}_D[\text{gap}] \le \mathbb{E}_D[\text{gap}^+] = \int_0^\infty \Pr[\text{gap} \ge \varepsilon]\,d\varepsilon \le \int_0^\infty \min\{1, \delta(\varepsilon)\}\,d\varepsilon.$$

We now compute the integral by exchanging the roles of ε and δ. The map $\varepsilon \mapsto \delta(\varepsilon)$ is continuous and strictly decreasing on $[0,\infty)$, with $\delta(0) = 4N_{2n}(\mathscr{H}) > 1$ and $\delta(\varepsilon) \to 0$ as $\varepsilon \to \infty$. Hence, for every $t \in (0,1)$ there is a unique $\varepsilon(t) >$

0 such that $\delta(\varepsilon(t)) = t$, and it is given by $\varepsilon(t) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ with $C = \sqrt{\log(4N_{2n}(\mathscr{H})/t)/n}$, by construction of $\delta(\varepsilon)$. Moreover, for $t \in (0,1)$,

$$\min\{1, \delta(\varepsilon)\} > t \iff \delta(\varepsilon) > t \iff \varepsilon < \varepsilon(t).$$

Writing $\min\{1, \delta(\varepsilon)\} = \int_0^1 \mathbf{1}\{t < \min\{1, \delta(\varepsilon)\}\}\, dt$ and exchanging the order of integration by Tonelli's theorem (the integrand is non-negative), we obtain

$$\int_0^\infty \min\{1, \delta(\varepsilon)\}\, d\varepsilon = \int_0^1 \int_0^\infty \mathbf{1}\{\varepsilon < \varepsilon(t)\}\, d\varepsilon\, dt = \int_0^1 \varepsilon(t)\, dt = \int_0^1 \varepsilon(\delta)\, d\delta.$$

Hence, writing $\varepsilon(\delta) = 4\sqrt{R^*_{\mathscr{H}}}\, C + 20C^2$ explicitly,

$$\mathbb{E}_D[\text{gap}] \le \int_0^1 \varepsilon(\delta)\, d\delta = 4\sqrt{\frac{R^*_{\mathscr{H}}}{n}} \int_0^1 \sqrt{\log \frac{4N_{2n}(\mathscr{H})}{\delta}}\, d\delta + \frac{20}{n} \int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta.$$

The logarithm is the only part depending on δ, and its integral is

$$\int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta = \log(4N_{2n}(\mathscr{H})) - \int_0^1 \log \delta\, d\delta = \log(4N_{2n}(\mathscr{H})) + 1.$$

Applying Jensen's inequality to the square root term,

$$\int_0^1 \sqrt{\log \frac{4N_{2n}(\mathscr{H})}{\delta}}\, d\delta \le \sqrt{\int_0^1 \log \frac{4N_{2n}(\mathscr{H})}{\delta}\, d\delta} = \sqrt{\log(4N_{2n}(\mathscr{H})) + 1},$$

we obtain

$$\mathbb{E}_D[\text{gap}] \le 4\sqrt{\frac{R^*_{\mathscr{H}}\,(\log(4N_{2n}(\mathscr{H})) + 1)}{n}} + 20\,\frac{\log(4N_{2n}(\mathscr{H})) + 1}{n}.$$

Asymptotically,

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le O\left(\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\, \log N_{2n}(\mathscr{H})}{n}} + \frac{\log N_{2n}(\mathscr{H})}{n}\right)$$

because, at the numerator,

$$\log(4N_{2n}(\mathscr{H})) + 1 = \log 4 + \log N_{2n}(\mathscr{H}) + 1 = \log N_{2n}(\mathscr{H}) + O(1)$$

and for large x,

$$\log x + O(1) \le C \log x$$

for some constant $C > 0$. Apply Lemma 3.7, obtaining

$$\mathbb{E}_D\left[(R - \hat{R}_D)(\hat{h}_D)\right] \le \min\left\{O\left(\sqrt{\frac{(A_{\mathscr{H}} + \sigma^2)\log|\mathscr{H}|}{n}} + \frac{\log|\mathscr{H}|}{n}\right), O\left(\sqrt{A_{\mathscr{H}} + \sigma^2} + 1\right)\right\}$$

□

Proof of Lemma 5.2. We prove the inequalities for $N_S(\mathscr{H})$, for every $S \in (\mathcal{X}\times\mathcal{Y})^n$. Since $N_D(\mathscr{H})$ takes the value $N_S(\mathscr{H})$ when D takes the value S, they hold for $N_D(\mathscr{H})$ for every realization of D.

Fix $S = ((x_1, y_1), \dots, (x_n, y_n)) \in (\mathcal{X} \times \mathcal{Y})^n$, and consider the maps

$$\Psi\colon \mathscr{H} \to \mathcal{Y}^n, \quad h \mapsto \big(h(x_1), \dots, h(x_n)\big),$$

$$\lambda\colon \mathcal{Y}^n \to \{0,1\}^n, \quad (\hat{y}_1, \dots, \hat{y}_n) \mapsto \big(\ell(\hat{y}_1, y_1), \dots, \ell(\hat{y}_n, y_n)\big).$$

The map $\Phi\colon h \mapsto (\ell_h(x_1, y_1), \dots, \ell_h(x_n, y_n))$ of the proof of Lemma 3.7 factorizes as $\Phi = \lambda \circ \Psi$, hence $N_S(\mathscr{H}) = |\Phi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))|$. Moreover, by definition of growth function, $|\Psi(\mathscr{H})| \le \Pi_n(\mathscr{H})$.

$1 \le N_S(\mathscr{H})$. Since $\mathscr{H}$ is nonempty, $\Phi(\mathscr{H})$ is nonempty.

$N_S(\mathscr{H}) \le \Pi_n(\mathscr{H})$. Since the image of a set under any map has cardinality at most that of the set,

$$N_S(\mathscr{H}) = |\lambda(\Psi(\mathscr{H}))| \le |\Psi(\mathscr{H})| \le \Pi_n(\mathscr{H}).$$

$\Pi_n(\mathscr{H}) \le \min\{|\mathscr{H}|, 2^n\}$. Let $x_1, \dots, x_n \in \mathcal{X}$ and let Ψ be the corresponding map. As above, $|\Psi(\mathscr{H})| \le |\mathscr{H}|$, and $\Psi(\mathscr{H}) \subseteq \mathcal{Y}^n$ with $|\mathcal{Y}^n| = 2^n$, since $\mathcal{Y} = \{-1, 1\}$. Taking the supremum over $x_1, \dots, x_n$ gives $\Pi_n(\mathscr{H}) \le |\mathscr{H}|$ and $\Pi_n(\mathscr{H}) \le 2^n$.

$\Pi_n(\mathscr{H}) = N_n(\mathscr{H})$. Taking the supremum over S in $N_S(\mathscr{H}) \le \Pi_n(\mathscr{H})$ gives $N_n(\mathscr{H}) \le \Pi_n(\mathscr{H})$. For the converse, let $x_1, \dots, x_n \in \mathcal{X}$, take any $y_1, \dots, y_n \in \mathcal{Y}$, and let $S = ((x_1, y_1), \dots, (x_n, y_n))$. Since $\mathcal{Y} = \{-1, 1\}$ and ℓ is the zero-one loss, $\hat{y}_i = y_i$ if $\ell(\hat{y}_i, y_i) = 0$ and $\hat{y}_i = -y_i$ if $\ell(\hat{y}_i, y_i) = 1$. Hence λ is injective, and

$$|\Psi(\mathscr{H})| = |\lambda(\Psi(\mathscr{H}))| = N_S(\mathscr{H}) \le N_n(\mathscr{H}).$$

Taking the supremum over $x_1, \dots, x_n$ gives $\Pi_n(\mathscr{H}) \le N_n(\mathscr{H})$. □

Appendix B. Preliminary experiments

B.1. **Noise structure.** We generate the labels with a centered parabola (see Section B.2), so that roughly half of them are 1 and half -1 in the interval $[0, 10]$. Then, we inspect what happens if we use a Bernoullian noise instead of a Gaussian one, i.e. when data is generated as

$$y \leftarrow sign(h^*(x))b$$

for $b \sim \text{Bernoulli}(p)$. Notice that this noise flips the labels with a probability of $1 - p$. While from the first experiments it seemed that the algorithm achieved a higher test error, we postulate that this may be due to the portion of misclassified inputs, rather than to the noise structure itself. Hence, fixing $p = 0.8$, we look for the value of σ that achieves roughly the same portion of mislcassified inputs (20%). Figure 8 shows that this happens for $\sigma = 5.5$, and Figures 9 and 10 show that the two noise structures then yield essentially the same train and test errors.

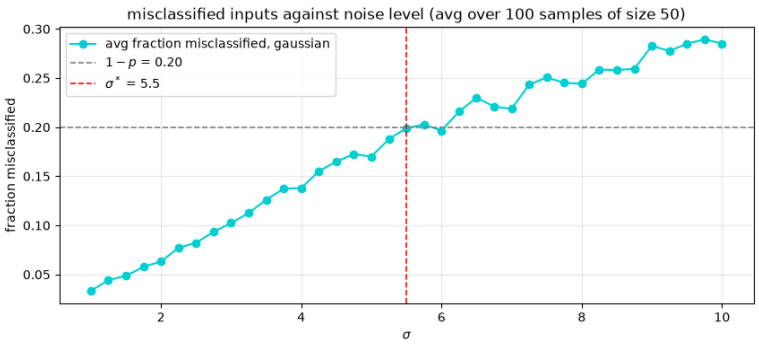


Figure 8. Calibrating Gaussian noise on the parabola. Average fraction of misclassified inputs against σ, over 100 samples of size $n = 50$.

B.2. **Data-generating function.** Our first choice of h^* is a parabola centered horizontally at 5, the midpoint of the support $\mathcal{X}$, and vertically so that roughly half of the points fall above the x-axis and half below. I.e.,

$$h^*(x) = (x - 5)^2 - 6.25$$

Figure 11 shows the data generated with the centered parabola, for different amounts of noise.

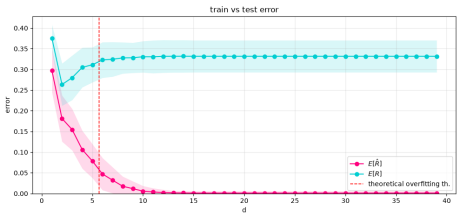


FIGURE 9. Parabola, Bernoulli noise with $p = 0.8$; $n = 50$, $N = 10,000$, $n_{\text{test}} = 200,000$.

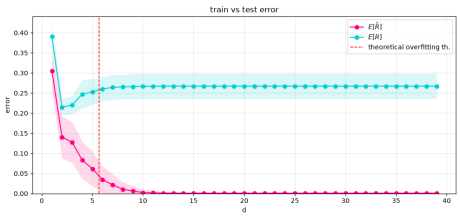


FIGURE 10. Parabola, Gaussian noise with $\sigma = 5.5$; $n = 50$, $N = 10,000$, $n_{\text{test}} = 200,000$.

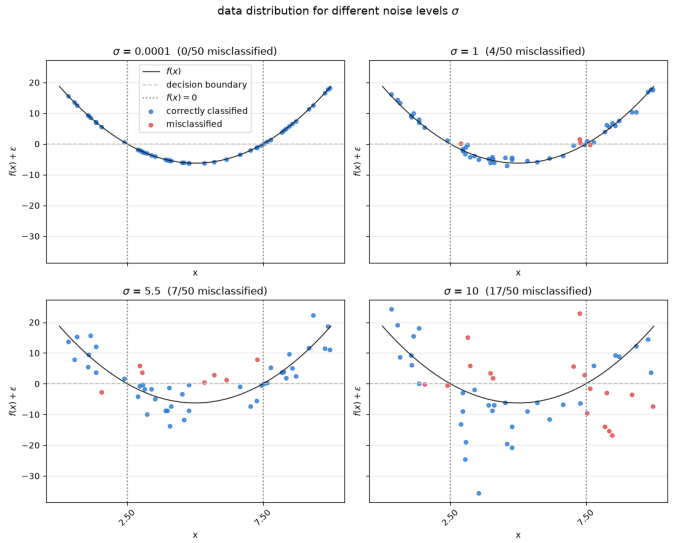


FIGURE 11. Samples from the parabola. One sample of size $n = 50$ for $\sigma \in \{10^{-4}, 1, 5.5, 10\}$.

Because in theory an algorithm could learn such a classification pattern as soon as it sees 3 points (one point for each region), we investigate what happens when the underlying function is more complex. We study a sinusoid with varying frequency,

$$h^*(x) = 5 \sin\left(\frac{\pi\,(m+1)}{10}\,x\right)$$

We multiply the sine by 5 to allow for enough height to properly inspect the data, while the argument of the sine is such that m is the number of times h^* crosses the x-axis in the open interval $(0, 10)$, yielding $m + 1$ contiguous blocks of $1s$ and $-1s$.

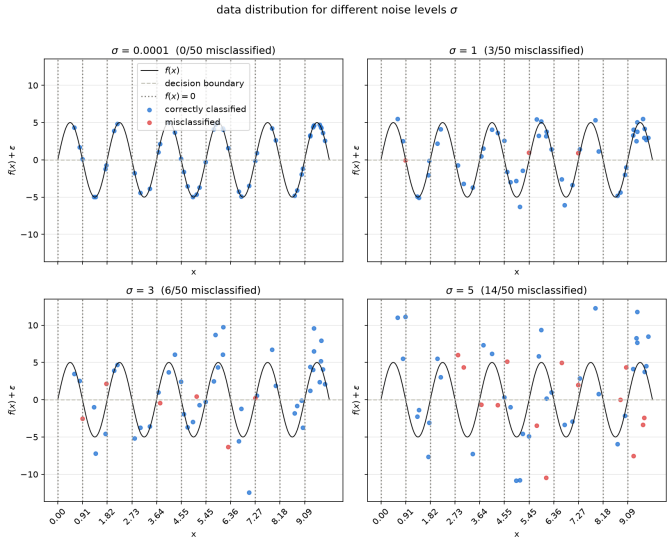


FIGURE 12. Samples from the sine ($m = 10$). One sample of size $n = 50$ for $\sigma \in \{10^{-4}, 1, 3, 5\}$.

Figure 12 shows the data generated with the sine for $m = 10$, for different amounts of noise.

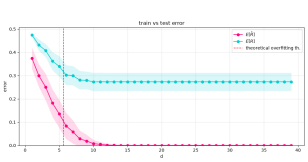


FIGURE 13. Sine, $m = 10$.

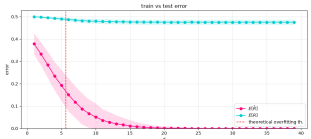


FIGURE 14. Sine, $m = 100$.

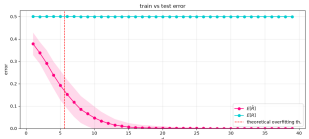


FIGURE 15. Sine, $m = 1000$.

We explore how a random sample and the performance of the algorithm change for different sine frequencies. In particular, we expect that for a high frequency the algorithm will not learn, as our datasize is $n = 50$. In Figures 13–15, we test a function that osciallets $10, 100$, and 1000 times with a noise level of $\sigma = 3$ (we reduce it compared to $\sigma = 5$, having increased the complexity of the function). We notice that for $m = 1000$ the algorithm fails to learn, and the test error stays stable at 0.5. This is because the labels change so frequently with respect to the number of points in the training dataset n that it is as if we had random labels. Hence, we

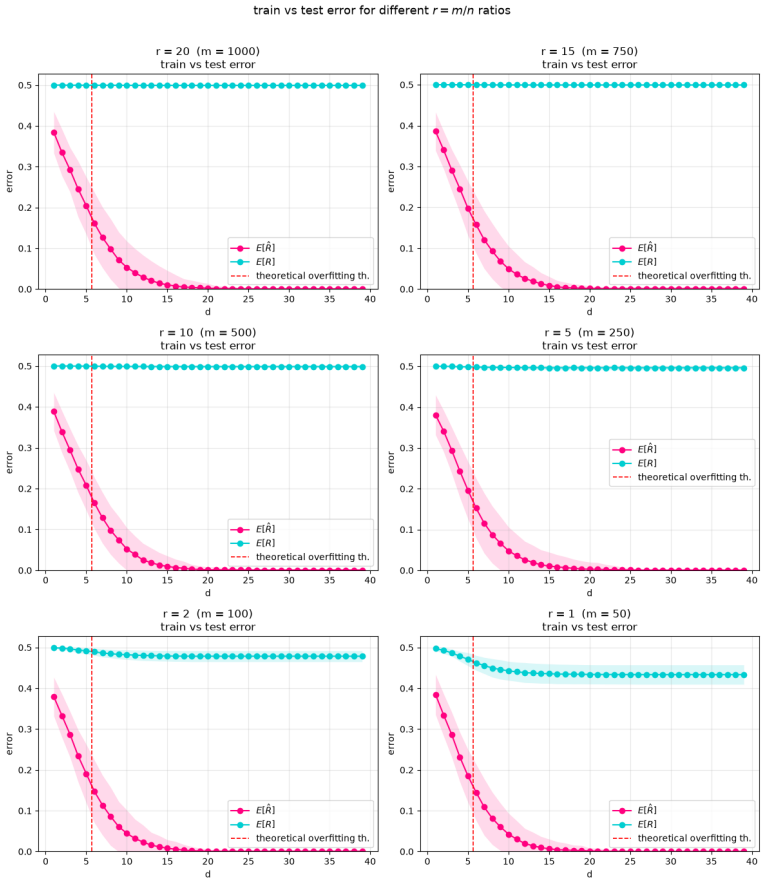


FIGURE 16. Learnability threshold in m/n. Train and test error for different ratios $r = m/n$; $n = 50$, $\sigma = 3$, $N = 10,000$, $n_{\text{test}} = 200,000$.

investigate the relation between the complexity of the underlying function m and the data size n. We find that for $\frac{m}{n} > 5$, roughly, the algorithm ceases to learn (Figure 16). In the following experiments, we fix $m = 10$ and $n = 50$, which is very far from the threshold.

We then verify whether a Gaussian noise with $\sigma = 5.5$ achieves the same portion of misclassified inputs on average as a Bernoullian noise with $p = 0.8$, as was the case for data generated with a parabola. Figure 17 shows that this is not the case on average (29% against 20%), while Figure 18 shows that the desired value is $\sigma = 3.5$.

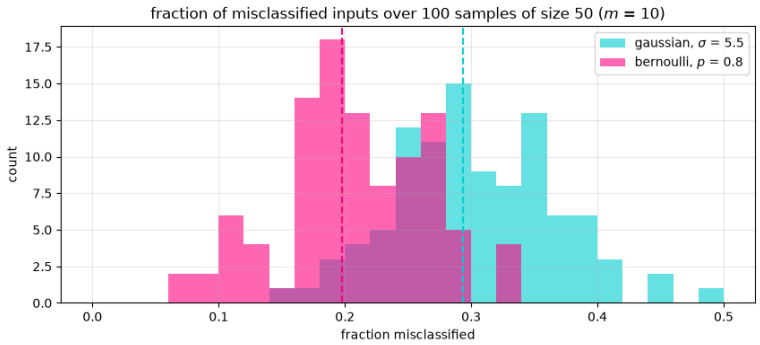


FIGURE 17. Gaussian against Bernoulli noise on the sine ($m = 10$). Fraction of misclassified inputs over 100 samples of size $n = 50$; dashed lines are the averages.

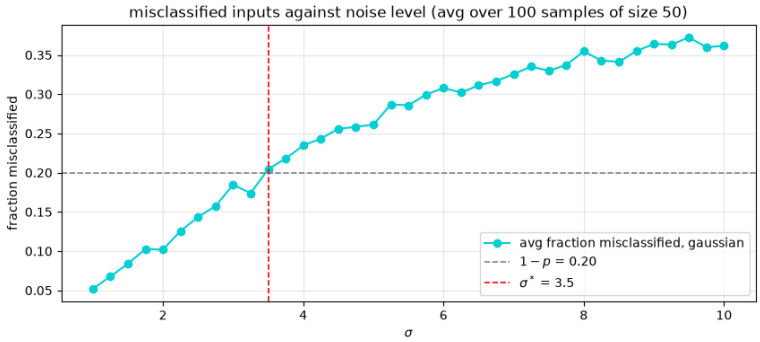


FIGURE 18. Calibrating Gaussian noise on the sine ($m = 10$). Average fraction of misclassified inputs against σ, over 100 samples of size $n = 50$.

Finally, we investigate the afore-mentioned intuition about how easy it is for the algorithm to learn a simple function. We take $m = 2$ (which corresponds to a parabola, in terms of classification pattern), sample uniformly from $\mathcal{X}$, and look for the training set size n for which the test error gets within 0.01 of the Bayes risk on average, at depth 2 and 3.

We start with little noise, $\sigma = 1$, for which the Bayes risk is ≈ 0.05 (Figure 19). Here the algorithm learns the function quickly: with $n = 50$ points the test error is already ≈ 0.07, within 0.02 of the Bayes risk, and it gets within 0.01 with $n = 200$ at depth 2 and $n = 500$ at depth 3.

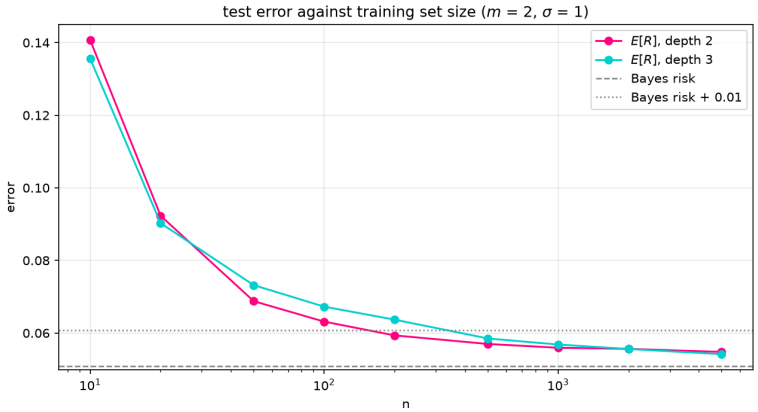


FIGURE 19. Sample size needed for a simple function, $\sigma = 1$. Test error at depth 2 and 3 against n for the sine with $m = 2$, averaged over 100 samples; $N = 10,000$, $n_{\text{test}} = 200,000$.

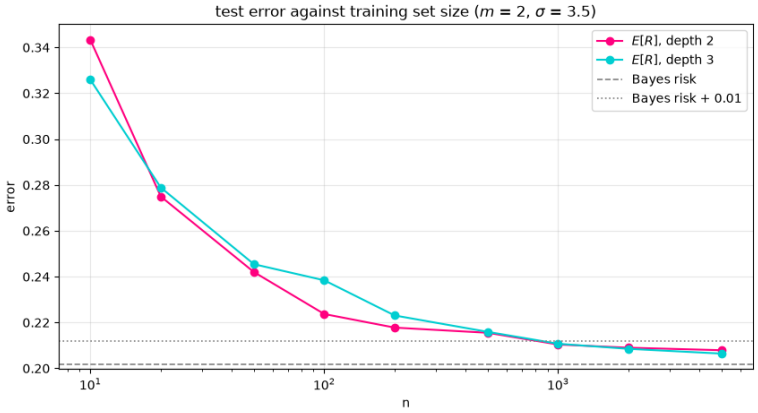


FIGURE 20. Sample size needed for a simple function, $\sigma = 3.5$. Test error at depth 2 and 3 against n for the sine with $m = 2$, averaged over 100 samples; $N = 10,000$, $n_{\text{test}} = 200,000$.

With our noise level, $\sigma = 3.5$, the Bayes risk is ≈ 0.20 and $n = 1000$ training points are needed on average for the test error to get within 0.01 of it, both at depth 2 and 3 (Figure 20). Hence, although two splits are enough to represent the function, with this amount of noise the algorithm needs many points to place them correctly.

Appendix C. Additional results

Fact C.1 (Bound on the Rademacher complexity of $\mathscr{L}$)**.** *Let ℓ take values in $[0, M]$ for some $M > 0$. Then, for any distribution p, any $\mathscr{H}$ and any $n \geq 1$,*

$$\mathcal{R}_n(\mathscr{L}) \leq \frac{M}{2}.$$

In particular, if ℓ is the zero-one loss, $\mathcal{R}_n(\mathscr{L}) \leq \frac{1}{2}$.

Proof. Fix a realization of D and ξ, and let $I_+ = \{i : \xi_i = 1\}$. For every $\ell_h \in \mathscr{L}$, since $0 \leq \ell_h(X_i, Y_i) \leq M$,

$$\frac{1}{n}\sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i) = \frac{1}{n}\sum_{i \in I_+} \ell_h(X_i, Y_i) - \frac{1}{n}\sum_{i \notin I_+} \ell_h(X_i, Y_i) \leq \frac{1}{n}\sum_{i \in I_+} \ell_h(X_i, Y_i) \leq \frac{M|I_+|}{n}.$$

The right-hand side does not depend on h, hence

$$\sup_{\ell_h \in \mathscr{L}} \frac{1}{n}\sum_{i=1}^{n} \xi_i \ell_h(X_i, Y_i) \leq \frac{M}{n}\sum_{i=1}^{n} \mathbf{1}(\xi_i = 1).$$

Taking expectations over ξ and D, and since $\Pr[\xi_i = 1] = \frac{1}{2}$,

$$\mathcal{R}_n(\mathscr{L}) \leq \frac{M}{n}\sum_{i=1}^{n} \Pr[\xi_i = 1] = \frac{M}{2}.$$

The zero-one loss takes values in $\{0, 1\}$, so the second claim follows with $M = 1$. □

Appendix D. Code specifications

Computation of the best risk. The algorithm is explained using one-based indexing for the sake of exposition, while the real code uses zero-based indexing as usual in Python. We produce N noiseless labels for each input value in the support and we cluster the intervals over $\mathcal{X}$ where inputs have the same label. Let L be the number of contiguous blocks we obtain. We create a `dp` matrix with dimension $k_{\max} \times 2$: in the columns, we have the labels $y_j \in \{1, -1\}$ for $j = 1, 2$; in the rows, we have the number of leaves in the tree $k = 1, \dots, k_{\max}$, with $k_{\max} = \min(2^{d_{\max}}, L)$ the maximum number of leaves and $d_{\max}$ the largest depth we consider overall (that is a multiple of $\lceil \log_2 n \rceil$ as mentioned in Section 2). Note that the tree reproducing exactly the noiseless labels has L leaves and has expected risk equal to $1 - p$, the portion of corrupted labels, reaching the Bayes risk. Since the best risk is bounded from below by the Bayes risk, considering more than L leaves cannot improve it. So, to avoid useless computations, we consider the minimum between L and $2^{d_{\max}}$. Given the true labels for each of the L blocks, it is easy to create also a `cost` matrix with dimension $L \times 2$, where `cost[i, j]` is the risk given by predicting y_j at block i. Let y_i be the true label in block i and ρ_i the portion of inputs falling into block i. We have

$$\texttt{cost[i,j]} = \begin{cases} \rho_i \cdot (1 - p) & \text{if } y_i = y_j \\ \rho_i \cdot p & \text{if } y_i \neq y_j \end{cases}$$

We initialize the `dp` matrix setting all entries to ∞. For the first iteration over block $i = 1$, we set

$$\texttt{dp[1, :]} = \texttt{cost[1, :]}$$

For each iteration $i = 2, \dots, L$ we dynamically update the `dp` matrix with minimum risks. Dynamically updating means that in `dp[k,j]` we store the cumulative minimum risk of having predicted label y_j at the *previous* block $i-1$ with k leaves, and we create a new $k_{\max} \times 2$ matrix called `new` such that in `new[k,j]` we store the minimum risk of predicting label y_j at the current block i with k leaves. This minimum risk is computed as follows. Given that we predict y_j at block i, we choose between the tree that has predicted label y_j at block $i - 1$ using k leaves, i.e. that keeps the same label, and the tree that has predicted label $-y_j$ at block

$i-1$ using $k-1$ leaves, i.e. that switches labels. These two values are stored in the `dp` matrix, as said. Then, we add the cost of predicting label y_j in the current block i. This is summarized by the following line

$$\texttt{new[k,j]} = \min(\texttt{dp[k,j]}, \texttt{dp[k-1,3-j]}) + \texttt{cost[i,j]}$$

Note that $3-j$ swaps the indices 1 and 2, i.e. it stays for $-y_j$. Also, this code holds for $k > 1$, while for $k = 1$ the cost of switchiing is set to ∞, as it is not admissible. After passing over all $2k_{\max}$ entries of `new`, we update `dp = new` and pass to the next block $i+1$. The `dp` matrix will now contain the cumulative minimum risks for each possible label predicted in the last block i. Note that using an auxiliary matrix `new` is not strictly necessary, as we could update the `dp` matrix recursively, but using it eases understanding the logic. In the end, after passing over the last contiguous block, we obtain a matrix with the minimum risk for each number of leaves and each label assigned to the last block. We create a $k_{\max}$-long list `exact_k_risk` with the minimum value for each k,

$$\texttt{exact_k_risk[k]} = \min_{j \in \{1,2\}} \texttt{dp[k,j]}$$

Then, we create another $k_{\max}$-long list `at_most_k_risk` that stores the cumulative minimum for each possible *maximum* number of leaves,

$$\texttt{at_most_k_risk[k]} = \min_{r \in [1,k]} \texttt{exact_k_risk[r]}$$

Thus, the k-th entry of `at_most_k_risk` contains the best risk allowing at most k leaves. Finally, for each depth d we set $k_d = \min(2^d, L)$ and select the k_d-th entry in `at_most_k_risk`, yielding the best risk among trees with maximum depth d.